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p-TORSION FOR UNRAMIFIED ARTIN–SCHREIER COVERS OF CURVES

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ABSTRACT. Let $Y \to X$ be an unramified Galois cover of curves over a perfect field k of characteristic $p > 0$ with $\mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$, and let J_X and J_Y be the Jacobians of X and Y respectively. We consider the p-torsion subgroup schemes $J_X[p]$ and $J_Y[p]$, analyze the Galois-module structure of $J_Y[p]$, and find restrictions this structure imposes on $J_Y[p]$ (for example, as manifested in its Ekedahl–Oort type) taking $J_X[p]$ as given.

1. Introduction

Let k be a perfect field of characteristic $p > 0$ with algebraic closure $\overline{k}$, and let X be a smooth, proper, geometrically irreducible curve of genus g_X over k. Let $\pi : Y \to X$ be an unramified Galois covering with $G := \mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ and with Y geometrically irreducible. Writing g_Y for the genus of Y, the Riemann-Hurwitz formula for π says $2g_Y - 2 = p(2g_X - 2)$.

Let J_X and J_Y be the Jacobians of X and Y, and let $J_X[p]$ and $J_Y[p]$ be their p-torsion subgroup schemes. These are self-dual BT_1 group schemes of orders p^{2g_X} and p^{2g_Y} respectively. (See Section 2 for the definitions of "self-dual" and "BT_1".) Our goal is to describe relations between $J_Y[p]$ and $J_X[p]$. We first consider the G-module structure of $J_Y[p]$ and show that the latter is close to being G-free in a suitable sense. We then explore the restrictions this structure imposes on $J_Y[p]$ (for example, as manifested in its Ekedahl–Oort type) taking $J_X[p]$ as given. In particular, over a general perfect base field k, we deduce inequalities for the "arithmetic p-rank" of Y, namely $\dim_{\mathbb{F}_p} J_Y[p](k)$, and over an algebraically closed base field k, we deduce new restrictions on the isomorphism type of $J_Y[p]$ which in turn yield new parity restrictions on the a-number of Y.

Recall that a p-torsion group scheme G over k has a canonical decomposition

$$\mathcal{G} \cong \mathcal{G}_{\acute{e}t} \oplus \mathcal{G}_m \oplus \mathcal{G}_{ll}$$

into étale, multiplicative, and local-local parts. (See Section 2.) Let f_X and f_Y be the p-ranks of X and Y respectively (i.e., the dimensions over $\mathbb{F}_p$ of $J_X[p]_{\acute{e}t}(\overline{k})$ and $J_Y[p]_{\acute{e}t}(\overline{k})$). The Deuring–Shafarevich formula in this context says that $f_Y - 1 = p(f_X - 1)$. There is a well-known refinement of this taking into account the G

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action. Indeed, let $\mathbb{F}_p[G]$ be the group ring of G over $\mathbb{F}_p$. If k is algebraically closed, then we have isomorphisms of group schemes with G action:

$$J_Y[p]_{\acute{e}t} \cong \mathbb{Z}/p\mathbb{Z} \oplus \left((\mathbb{Z}/p\mathbb{Z})^{f_X-1} \otimes_{\mathbb{F}_p} \mathbb{F}_p[G]\right) \tag{1.1}$$

and

$$J_Y[p]_m \cong \mu_p \oplus \left((\mu_p)^{f_X-1} \otimes_{\mathbb{F}_p} \mathbb{F}_p[G]\right), \tag{1.2}$$

where G acts trivially on the factors $\mathbb{Z}/p\mathbb{Z}$ and μ_p. (This follows from [Nak85, Theorem 2] or [Cre84, Theorem 1.5] and Cartier duality. It may also be obtained from the Hochschild-Serre spectral sequences for π with coefficients in $\mathbb{Z}/p\mathbb{Z}$ and μ_p.)

1.1. **Filtrations from the G action and their associated gradeds.** Our first aim in this paper is to study all of $J_Y[p]$ (i.e., the local-local part as well as the étale and multiplicative parts) and to allow k to be a general perfect field of characteristic p (giving "structural" results analogous to (1.1) and (1.2) for general k). To state the result, we first define a certain subquotient $\mathcal{G}_X$ of $J_X[p]$.

Proposition 1.1. *The covering $\pi : Y \to X$ and the isomorphism $\mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ give rise to canonical homomorphisms of groups schemes $J_X[p] \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}$ and $\mu_P \hookrightarrow J_X[p]$. Define a (self-dual BT_1) group scheme $\mathcal{G}_X$ by the exact sequences*

$$0 \to \mathcal{G}_{X,\acute{e}t} \to J_X[p]_{\acute{e}t} \to \mathbb{Z}/p\mathbb{Z} \to 0, \tag{1.3}$$

$$0 \to \mu_p \to J_X[p]_m \to \mathcal{G}_{X,m} \to 0, \tag{1.4}$$

and the isomorphism

$$\mathcal{G}_{X,ll} \cong J_X[p]_{ll}.$$

Let $\gamma \in G = \mathrm{Gal}(Y/X)$ be the element corresponding to $1 \in \mathbb{Z}/p\mathbb{Z}$ under the fixed isomorphism $G \cong \mathbb{Z}/p\mathbb{Z}$ and let δ be the element $\gamma - 1$ in the group ring $k[G]$. Then $k[G] \cong k[\delta]/(\delta^p)$. Thus δ induces a nilpotent endomorphism of $J_Y[p]$, and the kernels and images of powers of δ on $J_Y[p]$ give two (generally distinct) p-step filtrations by self-dual BT_1 group schemes. The following result describes the associated graded objects of these two filtrations. (We use the standard notations $\mathcal{G}[\delta]$ and $\mathcal{G}/\delta$ to denote kernels and cokernels respectively.)

Theorem 1.2.

(1) *There are canonical isomorphisms*

$$\frac{\delta^i J_Y[p]_{\acute{e}t}}{\delta^{i+1} J_Y[p]_{\acute{e}t}} \cong \frac{J_Y[p]_{\acute{e}t}[\delta^{p+1-i}]}{J_Y[p]_{\acute{e}t}[\delta^{p-i}]} \cong \mathcal{G}_{X,\acute{e}t} \quad \text{for } i = 1, \ldots, p-1,$$

as well as exact sequences

$$0 \to \mathcal{G}_{X,\acute{e}t} \to J_Y[p]_{\acute{e}t}[\delta] \to \mathbb{Z}/p\mathbb{Z} \to 0 \tag{1.5}$$

and

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{\acute{e}t}/\delta \to \mathcal{G}_{X,\acute{e}t} \to 0. \tag{1.6}$$

Pull back by π induces a canonical isomorphism $J_X[p]_{\acute{e}t} \cong J_Y[p]_{\acute{e}t}[\delta]$ which identifies the exact sequences (1.5) and (1.3).

(2) *There are canonical isomorphisms*

$$\frac{\delta^i J_Y[p]_m}{\delta^{i+1} J_Y[p]_m} \cong \frac{J_Y[p]_m[\delta^{p+1-i}]}{J_Y[p]_m[\delta^{p-i}]} \cong \mathcal{G}_{X,m} \quad \text{for } i = 1, \dots, p-1,$$

as well as exact sequences

$$0 \to \mu_p \to J_Y[p]_m/\delta \to \mathcal{G}_{X,m} \to 0 \tag{1.7}$$

and

$$0 \to \mathcal{G}_{X,m} \to J_Y[p]_m[\delta] \to \mu_p \to 0. \tag{1.8}$$

Push forward by π induces a canonical isomorphism $J_Y[p]_m/\delta \cong J_X[p]_m$ which identifies the exact sequences (1.7) *and* (1.4).

(3) *We have equalities*

$$\delta^i J_Y[p]_{ll} = J_Y[p]_{ll}[\delta^{p-i}]$$

for $i = 0, \dots, p$, as well as canonical isomorphisms

$$\frac{\delta^i J_Y[p]_{ll}}{\delta^{i+1} J_Y[p]_{ll}} \cong \frac{J_Y[p]_{ll}[\delta^{p-i}]}{J_Y[p]_{ll}[\delta^{p-i-1}]} \cong J_X[p]_{ll} \quad \text{for } i = 0, \dots, p-1.$$

As we will see in Section 4, the asymmetry between the kernels and cokernels of δ (i.e., (1.5) vs. (1.6) and (1.7) vs. (1.8)) is significant. Indeed, although $J_Y[p]_{\acute{e}t}[\delta]$ and $J_Y[p]_{\acute{e}t}/\delta$ have the same order and are closely related, they are not in general isomorphic. Similarly for $J_Y[p]_m[\delta]$ and $J_Y[p]_m/\delta$. Readers are referred to Figures 1 and 2 in Section 4 for pictorial versions of Theorems 1.2 and 1.5 in terms of Dieudonné modules. Among other things, the figures show how the two filtrations (by images and kernels of δ) interact.

When $k = \overline{k}$, we may recover the isomorphisms (1.1) and (1.2) from parts (1) and (2) of Theorem 1.2 using the fact that the category of p-torsion étale (resp. multiplicative) group schemes over k is semi-simple with unique simple object $\mathbb{Z}/p\mathbb{Z}$ (resp. μ_p). The situation for the local-local part is much more complicated even when k is algebraically closed and will be discussed in more detail below.

We now consider a certain freeness property of p-torsion group schemes with G action.

Lemma 1.3. *Let $\mathcal{G}$ be a finite commutative group scheme over k killed by p and equipped with an action of $G = \mathbb{Z}/p\mathbb{Z}$, i.e., a group scheme equipped with the structure of a module over $\mathbb{F}_p[G]$. We say $\mathcal{G}$ is G-*free *if the following equivalent conditions are satisfied:*

(1) *the Dieudonné module $M(\mathcal{G})$ is free over the group ring $k[G]$*
(2) $\mathcal{G}[\delta]/\delta^{p-1}\mathcal{G} = 0$
(3) $\mathcal{G}[\delta^{p-1}]/\delta\mathcal{G} = 0$
(4) *δ^{p-1} induces an isomorphism $\mathcal{G}/\delta \xrightarrow{\sim} \mathcal{G}[\delta]$*

Corollary 1.4. *$J_Y[p]_{ll}$ is G-free.*

Proof. Conditions (2), (3), and (4) in Lemma 1.3 follow immediately from part (3) of Theorem 1.2. The corollary also follows from Theorem 1.5. □

There is an elegant, uniform variant of Theorem 1.2 provided that X has a k-rational point.

Theorem 1.5. *Suppose that X has a k-rational point S, and let $T = \pi^{-1}(S)$ viewed as a closed subscheme of Y. Then there is a self-dual BT_1 group scheme $\mathcal{H}$ equipped with the structure of a module over $\mathbb{F}_p[G]$ with the following properties:*

(1) *$\mathcal{H}$ is G-free in the sense of Lemma 1.3.*
(2) *There are equalities $\delta^i\mathcal{H} = \mathcal{H}[\delta^{p-i}]$ for $i = 1, \dots, p$, as well as canonical isomorphisms*

$$\frac{\delta^i\mathcal{H}}{\delta^{i+1}\mathcal{H}} \cong \frac{\mathcal{H}[\delta^{p-i}]}{\mathcal{H}[\delta^{p-i-1}]} \cong J_X[p] \quad \textit{for } i = 0, \dots, p-1.$$

(3) *There are canonical exact sequences*

$$0 \to J_Y[p]_{\acute{e}t} \to \mathcal{H}_{\acute{e}t} \to \operatorname{Res}_{T/S} \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0,$$

and

$$0 \to \mu_p \to \operatorname{Res}_{T/S} \mu_p \to \mathcal{H}_m \to J_Y[p]_m \to 0,$$

and a canonical isomorphism

$$\mathcal{H}_{ll} \cong J_Y[p]_{ll}.$$

The group scheme $\mathcal{H}$ depends on the choice of S in an interesting way, see Section 7.1.

Theorems 1.2 and 1.5 identify the minimal subquotients of $J_Y[p]$ and $\mathcal{H}$ as $\mathbb{F}_p[G]$-modules, and one might hope to "reassemble" the group schemes from this information. However, the category of BT_1 group schemes is not well-behaved with respect to extensions (even when k is algebraically closed), so even taking $J_X[p]$ as known, the structure of the repeated extensions $J_Y[p]$ and $\mathcal{H}$ can be quite intricate. For example, in Section 6 we show that there can be no Jordan-Hölder theorem for BT_1 group schemes, and we show that there may be infinitely many non-isomorphic $\mathbb{D}_k[G]$-modules giving rise to a fixed isomorphism class of $\mathbb{D}_k$-module and a fixed isomorphism class of $k[G]$-module. Nevertheless, we are able to deduce significant restrictions on $J_Y[p]$.

1.2. Analysis of the étale part of $J_Y[p]$.

We now consider freeness and related splitting questions for the étale part of $J_Y[p]$. Similar results hold for the multiplicative part by Cartier duality, and we leave it to the reader to make them explicit.

Note that equation (1.1) implies that when k is algebraically closed, $J_Y[p]_{\acute{e}t}$ is the direct sum of $\mathbb{Z}/p\mathbb{Z}$ and a G-free group scheme. The following result gives criteria for the same structural result to hold over a general k.

Proposition 1.6.

(1) *The exact sequence* (1.5) *splits if and only if there is an exact sequence of k-group schemes*

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{\acute{e}t} \to \mathcal{Q} \to 0,$$

where $\mathcal{Q}$ is G-free.

(2) *The exact sequence* (1.6) *splits if and only if there is an exact sequence of k-group schemes*

$$0 \to \mathcal{K} \to J_Y[p]_{ét} \to \mathbb{Z}/p\mathbb{Z} \to 0,$$

where $\mathcal{K}$ is G-free.

(3) *The exact sequences* (1.5) *and* (1.6) *both split if and only if $J_Y[p]_{ét}$ is the direct sum of $\mathbb{Z}/p\mathbb{Z}$ and a G-free group scheme.*

We will see in Example 7.1 that (1.5) and (1.6) may or may not split, and splitting of one does not in general imply splitting of the other; similarly for (1.7) and (1.8).

For a commutative p-torsion group scheme $\mathcal{G}$ over k, define the *arithmetic p-rank of $\mathcal{G}$*, denoted $\nu(G)$, by

$$p^{\nu(\mathcal{G})} = |\mathcal{G}(k)| = |\mathcal{G}_{ét}(k)|.$$

Let $\nu_X = \nu(J_X[p])$ and $\nu_Y = \nu(J_Y[p])$.

We say that $J_X[p]_{ét}$ (resp. $J_Y[p]_{ét}$) is *completely split* over k if $\nu_X = f_X$ (resp. $\nu_Y = f_Y$), or equivalently, if $J_X[p]_{ét} \cong (\mathbb{Z}/p\mathbb{Z})^{f_X}$ (resp. if $J_Y[p]_{ét} \cong (\mathbb{Z}/p\mathbb{Z})^{f_Y}$ as group schemes ignoring the G action). It follows from Theorem 1.2 that $J_Y[p]_{ét}$ is completely split over k if and only if we have an isomorphism of group schemes with G action as in equation (1.1).

We have the following general results on the arithmetic p-ranks of X and Y.

Theorem 1.7.

(1) *We have $\nu_X \le \nu_Y \le p\nu_X$.*
(2) *If the exact sequence* (1.3) (*equivalently* (1.5)) *splits, then we have the stronger upper bound $\nu_Y - 1 \le p(\nu_X - 1)$.*
(3) *If k is finite or algebraically closed, then $\nu_X \ge 1$.*
(4) *If k is finite,* (1.5) *is split, and* (1.6) *is non-split, then $\nu_X \ge 2$.*

See Example 7.3 for a discussion of the surprising possibility that ν_X might be 0 if k is infinite.

We also have bounds on the degrees of extensions of k over which certain splitting behaviors occur:

Theorem 1.8.

(1) *There is a finite Galois extension k' of k of exponent dividing p such that the sequences* (1.3) *and* (1.5) *split over k'. Thus* (*by Proposition* 1.6), *over k', $J_Y[p]_{ét}$ is an extension of G-free group scheme by $\mathbb{Z}/p\mathbb{Z}$.*
(2) *If $J_X[p]_{ét}$ is completely split then there is a finite Galois extension k' over k with exponent dividing p such that $J_Y[p]_{ét}$ is completely split over k'.*

In particular, if k is finite, the splitting behavior in Theorem 1.8 happens over extensions k'/k of degree dividing p.

A splitting not discussed in Theorem 1.8 (going from sequences (1.3) and (1.5) being split to $J_X[p]_{ét}$ being completely split) is controlled by $\mathrm{GL}_{f_X-1}(\mathbb{F}_p)$, and unfortunately, the exponent of this group grows rapidly with f_X.

We can give a very complete description of the group scheme $J_Y[p]_{ét}$ when k is finite and $f_X \le 2$.

Theorem 1.9.

(A) *If* $f_X = 1$, *then* $J_Y[p]_{ét} \cong J_X[p]_{ét} \cong \mathbb{Z}/p\mathbb{Z}$.

(B) *Suppose that* $p > 2$, k *is finite, and* $f_X = 2$. *Then exactly one of the following holds:*

(1a) $\nu_X = \nu_Y = 1$. *In this case, there is an isomorphism*

$$J_Y[p]_{ét} \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathcal{G},$$

where $\mathcal{G}$ *is* G*-free of rank 1 and* $\nu(\mathcal{G}) = 0$. *There is a non-trivial extension* k' *of* k *of degree dividing* $p-1$ *such that* $|\mathcal{G}(k')| = p^\mu$ *where* $1 \leq \mu \leq p$, *and there is an extension* k'' *of* k *of degree dividing* p *such that* $\mathcal{G} \cong (\mathcal{G}')^p$ *over* k'' *where* $\mathcal{G}'$ *has rank 1 and* $\nu(\mathcal{G}') = 0$. *Over* $k'k''$, $J_Y[p]_{ét}$ *is completely split.*

(1b) $\nu_X = 1$ *and* $\nu_Y > 1$. *In this case,* $\nu_Y < p+1$ (*so* $J_Y[p]_{ét}$ *is not completely split over* k), *and* $J_Y[p]_{ét}$ *is completely split over the extension of* k *of degree* p.

(2) $\nu_X = 2$. *In this case,* $2 \leq \nu_Y \leq p+1$ *and* $J_Y[p]_{ét}$ *is completely split over an extension of degree dividing* p.

(C) *Suppose that* $p = 2$, k *is finite, and* $f_X = 2$. *Then exactly one of the following holds:*

(1) $\nu_X = 1$. *In this case,* $1 \leq \nu_Y < 3$ (*so* $J_Y[p]_{ét}$ *is not completely split over* k), *and* $J_Y[p]_{ét}$ *splits completely over an extension of* k *of degree dividing 4.*

(2) $\nu_X = 2$. *In this case,* $2 \leq \nu_Y \leq 3$, *there is an exact sequence*

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{ét} \to \mathcal{Q} \to 0,$$

where $\nu(\mathcal{Q}) \geq 1$, *and* $J_Y[p]_{ét}$ *splits completely over an extension of* k *of degree dividing* p.

See the proof in Section 7 for additional information on the structure of $J_Y[p]_{ét}$ in each case.

1.3. **Analysis of the local-local part of** $J_Y[p]$. For the local-local part, it seems hopeless to give a full analysis, even when k is algebraically closed. Nevertheless, certain cases can be described rather explicitly. Recall that the a-number of a finite group scheme $\mathcal{J}$ over k is the largest integer a such that there is an injection $\alpha_p^a \hookrightarrow \mathcal{J}$. (When $\mathcal{J}$ is local-local, the a-number also has an interpretation as the number of generators and relations of the Dieudonné module of $\mathcal{J}$.) Write a_X and a_Y for the a-numbers of $J_X[p]$ and $J_Y[p]$ respectively. Booher and Cais [BC20, §6E] have observed that in our context, $a_X \leq a_Y \leq pa_X$. We can improve and refine this in some cases.

We say that a BT_1 group scheme is "superspecial" if it is isomorphic to a power of $E_{ss}[p]$ where E_{ss} is a supersingular elliptic curve.

Theorem 1.10. *Assume that* $p > 2$ *and that* k *is algebraically closed.*

(1) *If* $f_X = g_X - 1$ (*which implies that* $a_X = 1$ *and* $J_X[p]_{ll} \cong E_{ss}[p]$), *then*

$$a_Y \in \{2, 4, \ldots, p-1, p\}.$$

Moreover the local-local part of $J_Y[p]$ *has an explicit description in terms of generators and relations depending only on* a_Y.

(2) *More generally, if the local-local part of* $J_X[p]$ *is superspecial and* $a_Y < p$, *then* a_Y *is even.*

(3) *If $J_X[p]_{ll}$ is superspecial and $a_Y = pa_X$ or $pa_X - 1$ then there are few possibilities for the local-local part of $J_Y[p]$, and they can all be described explicitly in terms of generators and relations.*

Part (1) is proven as Theorem 8.7, and the precise description in terms of generators and relations is given there using Dieudonné modules. Part (2) and other features of the case where $J_X[p]_{ll}$ is superspecial are given in Theorem 8.10, and part (3) is proven as Theorem 8.11.

When $p = 2$, we can completely analyze the possibilities for $J_Y[p]_{ll}$ when $J_X[p]_{ll}$ is superspecial. They can be stated in terms of Ekedahl–Oort structures or multisets of words on the alphabet $\{f, v\}$. (See [Oor01] or [PU21] for background.)

Theorem 1.11. *Suppose that $p = 2$ and that $J_X[p]_{ll}$ is superspecial. Then the Ekedahl–Oort structure of $J_Y[p]_{ll}$ starts with h zeroes, i.e., it has the form $[0, 0, \dots, 0, \psi_{h+1}, \dots, \psi_{ph}]$. The multiset of words associated to $J_Y[p]_{ll}$ consists of fv with even multiplicity and of words obtained by concatenating words of the form $f^2(vf)^{e_1}v^2(fv)^{e_2}$ with $e_1, e_2 \geq 0$.*

Theorem 1.11 reduces the number of possibilities for $J_X[p]_{ll}$ from 2^{2h-1} to 2^h. It will be proven in Theorem 8.13 via a complete analysis of Dieudonné modules with the properties enjoyed by $J_Y[p]_{ll}$ in this context.

1.4. **Examples.** Throughout the paper, we report on numerous examples and counterexamples computed in Magma using code extending [Wei20].

1.5. **Outline of the paper.** We now describe the main outlines of the paper. By a theorem of Oda, the Dieudonné module of $J_Y[p]$ is isomorphic to the de Rham cohomology $H^1_{dR}(Y)$, so we are led to study the action of G on this and related cohomology groups. After discussing preliminaries on Dieudonné modules and $k[G]$-modules in Sections 2 and 3, we will prove crucial results of Chevalley-Weil type on the G-module structure of various flat and coherent cohomology groups in Section 4 and deduce the Dieudonné module version of Theorems 1.2 and 1.5. In particular, we show that $H^1_{dR}(Y)$ is close to being a free module over the group ring $k[G]$, and we control the Dieudonné structure of the "errors". See Propositions 4.1 and 4.3 for the precise statements. The translation from modules to groups is given in Section 5.

In Section 6, we review some of the difficulties in recovering $J_Y[p]$ from its associated graded as given in Theorem 1.2.

Our results on the étale part of $J_Y[p]$ (Proposition 1.6 and Theorems 1.7, 1.8, and 1.9) are proven in Section 7, mostly using the correspondence between étale group schemes and Galois representations. In Section 8, we axiomatize properties of the Dieudonné module of the local-local part of $J_Y[p]$, introduce a useful set of coordinates, and then prove the results underlying Theorems 1.10 and 1.11.

1.6. **Standing notation.** We fix the following notation and hypotheses for the rest of the paper: k is a perfect field of characteristic $p > 0$ with algebraic closure $\overline{k}$; X is a smooth, proper, geometrically irreducible curve of genus g_X over k; $\pi : Y \to X$ is an unramified Galois covering with group $\mathbb{Z}/p\mathbb{Z}$ such that Y has genus g_Y and is geometrically irreducible; and we fix an isomorphism $G := \mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$.

2. Dieudonné theory

We will use Dieudonné theory to study $J_X[p]$ and $J_Y[p]$. We refer to [Fon77] for the basic facts recalled below.

Let $\mathbb{D}_k$ be the associative k-algebra generated by symbols F and V with relations

$$FV = VF = 0, \quad F\alpha = \alpha^p F, \quad \text{and} \quad \alpha V = V\alpha^p$$

for all $\alpha \in k$.

A *p-group scheme over k* is by definition a finite commutative group scheme over k which is annihilated by p.

Recall that Dieudonné theory gives a contravariant equivalence between the category of p-group schemes over k and the category of left $\mathbb{D}_k$-modules of finite length. If G is a p-group scheme over k, we write $M(G)$ for the corresponding $\mathbb{D}_k$-module.

By definition, a $\mathbb{D}_k$-module M is *self-dual* if it admits a non-degenerate pairing of $\mathbb{D}_k$-modules, i.e., a non-degenerate k-bilinear pairing $\langle\cdot,\cdot\rangle$ with the properties that

$$\langle Fx, y\rangle = \langle x, Vy\rangle^p \quad \text{and} \quad \langle Vx, y\rangle = \langle x, Fy\rangle^{1/p} \tag{2.1}$$

for all $x, y \in M$. By definition, a $\mathbb{D}_k$-module M is a BT_1 *module* if $\operatorname{Ker}(F) = \operatorname{Im}(V)$ (or equivalently, if $\operatorname{Im}(F) = \operatorname{Ker}(V)$).

By definition, a p-group scheme G over k is *self-dual* (resp. a BT_1 *group scheme*) if $M(G)$ is self-dual (resp. a BT_1 module).

Any finite-dimensional $\mathbb{D}_k$-module N decomposes as

$$N \cong N_{ét} \oplus N_m \oplus N_{ll}, \tag{2.2}$$

where $N_{ét}$ is étale (F is bijective and $V = 0$), N_m is multiplicative ($F = 0$ and V is bijective), and N_{ll} is "local-local" (F and V are nilpotent). (Choose a sufficiently large integer a and set $N_{ét} = \operatorname{Im} F^a$, $N_m = \operatorname{Im} V^a$, and $N_{ll} = \operatorname{Ker} F^a \cap \operatorname{Ker} V^a$.)

The assignments $N \rightsquigarrow N_{ét}, N_m, N_{ll}$ are exact functors on the category of $\mathbb{D}_k$-modules. We denote the corresponding functors on p-torsion group schemes by $\mathcal{G} \rightsquigarrow \mathcal{G}_{ét}, \mathcal{G}_m$ and $\mathcal{G}_{ll}$.

If N is a $\mathbb{D}_k[G]$-module, the decomposition is also respected by the G action since G commutes with F and V. Also, N is self-dual if and only if $N_{ét}$ is dual to N_m and N_{ll} is self-dual; and N is a BT_1 module if and only if N_{ll} is a BT_1 module.

A theorem of Oda [Oda69, Cor. 5.11] says that for a smooth, proper, irreducible curve Z over k with Jacobian J_Z, the p-torsion subgroup $J_Z[p]$ is a self-dual BT_1 group scheme, and $M(J_Z[p]) \cong H^1_{dR}(Z)$ where $H^1_{dR}(Z)$ is equipped with a natural $\mathbb{D}_k$-module structure.[1] We will use this result to prove Theorems 1.2 and 1.5 as statements about $H^1_{dR}(X)$ and $H^1_{dR}(Y)$ and related Dieudonné modules.

3. G-modules

Consider the group rings $k[G]$ or $\mathbb{F}_p[G]$ where as before $G = \operatorname{Gal}(Y/X)$ and we have fixed an isomorphism $G \cong \mathbb{Z}/p\mathbb{Z}$. Let $\gamma \in G$ be the element corresponding to $1 \in \mathbb{Z}/p\mathbb{Z}$. Then

$$k[G] \cong k[\gamma]/(\gamma^p - 1) \cong k[\delta]/(\delta^p),$$

[1] Oda's result requires that Z have a k-rational point. This is of course no restriction when k is algebraically closed. When k is only assumed to be perfect, [Cai10, Prop. 5.4] shows that Oda's result continues to hold even without a rational point.

where $\delta := \gamma - 1$. Note that

$$\delta^{p-1} = \gamma^{p-1} + \cdots + 1$$

is the trace element of $k[G]$.

It is easily checked that up to isomorphism the indecomposable $k[G]$-modules are

$$V_i := k[\delta]/(\delta^i) \quad \text{for } i = 1, \ldots, p.$$

(See, e.g., [CR62, Lemma 64.2].) By the Krull–Schmidt theorem, every finitely generated $k[G]$-module is (non-canonically) isomorphic to a direct sum of indecomposable modules.

Lemma 3.1. *Let M be a finitely generated $k[G]$-module. Then the following conditions are equivalent:*

(1) *M is free over $k[G]$,*
(2) *$M[\delta] = \delta^{p-1}M$,*
(3) *$M[\delta^{p-1}] = \delta M$,*
(4) *δ^{p-1} induces an isomorphism $M/\delta M \cong M[\delta]$.*

Proof. This follows immediately from the classification of indecomposable $k[G]$-modules and a straightforward calculation. □

Equivalence of the conditions in Lemma 1.3 follows from Lemma 3.1 by applying the Dieudonné functor.

Define the dual of V_i as $V_i^\vee = \mathrm{Hom}_k(V_i, k)$ with action $(\gamma\phi)(v) = \phi(\gamma^{-1}v)$ for $\phi \in V_i^\vee$ and $v \in V_i$. Then $V_i^\vee \cong V_i$ (non-canonically) as $k[G]$-modules.

A non-degenerate, bilinear form $\langle \cdot, \cdot \rangle$ on M such that $\langle \gamma m, \gamma n \rangle = \langle m, n \rangle$ for all $m, n \in M$ induces an isomorphism $M \cong M^\vee$ of $k[G]$-modules.

Defining $\tilde{\delta} := \gamma^{-1} - 1 = -\gamma^{-1}\delta$, we have

$$\langle m, \delta n \rangle = \langle m, \gamma n \rangle - \langle m, n \rangle = \langle \gamma^{-1} m, n \rangle - \langle m, n \rangle = \langle \tilde{\delta} m, n \rangle,$$

for all $m, n \in M$. Note as well that δ and $\tilde{\delta}$ have the same image and kernel on any $k[G]$-module.

Parallel definitions and results hold for $\mathbb{F}_p[G]$-modules. We write W_j for the module $\mathbb{F}_p[\delta]/(\delta^j)$ over $\mathbb{F}_p[G] \cong \mathbb{F}_p[\delta]/(\delta^p)$.

4. de Rham cohomology as a $\mathbb{D}_k[G]$-module

Readers are assumed to be familiar with the flat, coherent, and de Rham cohomology of curves over perfect fields, and in particular with the semi-linear endomorphisms F and V of the de Rham cohomology of a curve. We recommend [Mil25], [MO15], and [Oda69] as basic references.

Suppose that Z is a smooth, proper, irreducible curve over k. Then we have coherent cohomology groups $H^s(Z, \mathcal{O}_Z)$ and $H^s(Z, \Omega^1_Z)$, as well as de Rham cohomology groups $H^s_{dR}(Z)$. These are finite-dimensional k vector spaces, and there is an exact sequence

$$0 \to H^0(Z, \Omega^1_Z) \to H^1_{dR}(Z) \to H^1(Z, \mathcal{O}_Z) \to 0. \tag{4.1}$$

There is a cup product on $H^1_{dR}(Z)$ which induces a perfect alternating pairing

$$H^1_{dR}(Z) \times H^1_{dR}(Z) \to H^2_{dR}(Z) = k$$

denoted $\langle \cdot, \cdot \rangle_Z$. The subspace $H^0(Z, \Omega^1_Z)$ is isotropic, and the pairing restricts to the (perfect) Serre duality pairing

$$H^0(Z, \Omega^1_Z) \times H^1(Z, \mathcal{O}_Z) \to k.$$

There are also semi-linear operators F and V on $H^s_{dR}(Z)$ making it into a $\mathbb{D}_k$-module. Explicitly,

$$H^0_{dR}(Z) \cong k, \text{ with } F\alpha = \alpha^p \text{ and } V\alpha = 0,$$

$$H^2_{dR}(Z) \cong k, \text{ with } F\alpha = 0 \text{ and } V\alpha = \alpha^{1/p},$$

in other words

$$H^0_{dR}(Z) \cong M(\mathbb{Z}/p\mathbb{Z}) \quad \text{and} \quad H^2_{dR}(Z) \cong M(\mu_p).$$

If (ω_i, f_{ij}) is a hypercocycle for an affine open cover $\{U_i\}$ of Z representing a class $c \in H^1_{dR}(Z)$, then Fc and Vc are represented by

$$(0, f_{ij}^p) \quad \text{and} \quad (\mathcal{C}\omega_i, 0) \tag{4.2}$$

respectively, where $\mathcal{C}$ is the Cartier operator. See [Oda69] for more details.

We have

$$\operatorname{Im}\left(V : H^1_{dR}(Z) \to H^1_{dR}(Z)\right) = \operatorname{Ker}\left(F : H^1_{dR}(Z) \to H^1_{dR}(Z)\right) = H^0(Z, \Omega^1),$$

so $H^1_{dR}(Z)$ is a BT_1 module. The pairing is compatible with the $\mathbb{D}_k$-module structure in the sense of equation (2.1), so $H^1_{dR}(Z)$ is a self-dual BT_1 module.

If π is a finite surjective map of smooth, projective curves, we have maps π^* and π_* on de Rham cohomology which are compatible with the $\mathbb{D}_k$-module structures. Also, if π is a Galois cover, $\pi^*\pi_*$ is the trace map. Applied to $\pi : Y \to X$, this means

$$\pi^*\pi_* = 1 + \gamma + \cdots + \gamma^{p-1} = \delta^{p-1} \tag{4.3}$$

as endomorphisms of de Rham cohomology.

With data $\pi : Y \to X$, $G = \operatorname{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ as usual, we will prove two results on the cohomology of X and Y (Propositions 4.1 and 4.3) which will yield Theorems 1.2 and 1.5.

Recall that the p-rank of a curve Z is by definition the integer f_Z such that $J_Z[p](\overline{k}) \cong (\mathbb{Z}/p\mathbb{Z})^{f_Z}$. It is also equal to the dimension over k of $H^1_{dR}(Z)_{ét}$.

Proposition 4.1.

(1) *There are canonical homomorphisms of $\mathbb{D}_k$-modules*

$$M(\mathbb{Z}/p\mathbb{Z}) \hookrightarrow H^1_{dR}(X)_{ét} \hookrightarrow H^1_{dR}(X) \quad \textit{and} \quad H^1_{dR}(X) \twoheadrightarrow H^1_{dR}(X)_m \twoheadrightarrow M(\mu_p)$$

which are exchanged by Cartier duality. The Dieudonné module of the group $\mathcal{G}_X$ in Proposition 1.1 is

$$\mathcal{M}_X := \frac{\operatorname{Ker}\left(H^1_{dR}(X) \twoheadrightarrow M(\mu_p)\right)}{\operatorname{Im}\left(M(\mathbb{Z}/p\mathbb{Z}) \hookrightarrow H^1_{dR}(X)\right)}.$$

(2) *There are (non-canonical) isomorphisms of $k[G]$-modules*

$$H^1_{dR}(Y)_{\acute{e}t} \cong V_1 \oplus V_p^{f_X-1},$$
$$H^1_{dR}(Y)_m \cong V_1 \oplus V_p^{f_X-1},$$
$$H^1_{dR}(Y)_{ll} \cong V_p^{2h_X},$$

and

$$H^1_{dR}(Y) \cong V_1^2 \oplus V_p^{2g_X-2},$$

where $h_X = g_X - f_X$.

(3) *π^* induces isomorphisms of $\mathbb{D}_k$-modules*

$$H^1_{dR}(X)_m \xrightarrow{\pi^*} H^1_{dR}(Y)_m[\delta] \quad \text{and} \quad H^1_{dR}(X)_{ll} \xrightarrow{\pi^*} H^1_{dR}(Y)_{ll}[\delta],$$

and an exact sequence

$$0 \to M(\mathbb{Z}/p\mathbb{Z}) \to H^1_{dR}(X)_{\acute{e}t} \xrightarrow{\pi^*} H^1_{dR}(Y)_{\acute{e}t}[\delta] \to M(\mathbb{Z}/p\mathbb{Z}) \to 0.$$

The image $\pi^*\left(H^1_{dR}(X)_{\acute{e}t}\right)$ is equal to $\delta^{p-1}H^1_{dR}(Y)_{\acute{e}t}$.

(4) *π_* induces an isomorphism*

$$H^1_{dR}(Y)_{\acute{e}t}/\delta H^1_{dR}(Y)_{\acute{e}t} \xrightarrow{\pi_*} H^1_{dR}(X)_{\acute{e}t}$$

which identifies the line

$$\begin{aligned}\operatorname{Im}&\left(H^1_{dR}(Y)_{\acute{e}t}[\delta] \to H^1_{dR}(Y)_{\acute{e}t}/\delta H^1_{dR}(Y)_{\acute{e}t}\right)\\ &= \operatorname{Ker}\left(H^1_{dR}(Y)_{\acute{e}t}/\delta H^1_{dR}(Y)_{\acute{e}t} \xrightarrow{\delta^{p-1}} H^1_{dR}(Y)_{\acute{e}t}[\delta]\right)\end{aligned}$$

with the line

$$\operatorname{Im}\left(M(\mathbb{Z}/pZ) \hookrightarrow H^1_{dR}(X)_{\acute{e}t}\right)$$

defined in part (1).

Remark 4.2.

(1) It is straightforward to check that the cup product on de Rham cohomology induces a duality between $H^1_{dR}(Y)_{\acute{e}t}$ and $H^1_{dR}(Y)_m$ and its restriction to $H^1_{dR}(Y)_{ll}$ is perfect and gives the latter the structure of a self-dual module. It is also easy to see that $H^1_{dR}(Y)_{ll}$ is a BT_1. Since the pairing satisfies the compatibility $\langle \delta m, n\rangle_Y = \langle m, \tilde{\delta} n\rangle_Y$ (see Section 3), we find that the orthogonal complement of $\delta^i H^1_{dR}(Y)_{ll}$ is $\delta^{p-i}H^1_{dR}(Y)_{ll}$. This implies that

$$\frac{H^1_{dR}(Y)_{ll}}{\delta^i H^1_{dR}(Y)_{ll}} \quad \text{is dual to} \quad H^1_{dR}(Y)_{ll}[\delta^i].$$

On the other hand, δ^{p-i} induces an isomorphism

$$\frac{H^1_{dR}(Y)_{ll}}{\delta^i H^1_{dR}(Y)_{ll}} \to H^1_{dR}(Y)_{ll}[\delta^i].$$

Therefore, each of the submodules $H^1_{dR}(Y)_{ll}[\delta^i]$ is self-dual, and they are easily seen to be BT_1 modules as well. (One slight subtlety: These pairings are not compatible with restriction. Indeed, the restriction to $H^1_{dR}(Y)_{ll}[\delta^{i-1}]$ of the pairing just constructed on $H^1_{dR}(Y)_{ll}[\delta^i]$ is degenerate for $i > 1$.)

(2) The maps in part (1) may also be interpreted as a 3-step filtration on $H^1_{dR}(X)$ with

$$\begin{aligned} H^1_{dR}(X)^3 &= H^1_{dR}(X), \\ H^1_{dR}(X)^2 &= \operatorname{Ker}\left(H^1_{dR}(X) \to M(\mu_p)\right), \\ H^1_{dR}(X)^1 &= \operatorname{Im}\left(M(\mathbb{Z}/p\mathbb{Z}) \to H^1_{dR}(X)\right), \end{aligned}$$

and

$$H^1_{dR}(X)^0 = 0.$$

The subquotients are $M(\mu_p)$, $\mathcal{M}_X$, and $M(\mathbb{Z}/p\mathbb{Z})$.

(3) The filtration above is self-dual in the sense that $H^1_{dR}(X)^2$ and $H^1_{dR}(X)^1$ are orthogonal complements of one another.

Figure 1 may help to digest the statement of the Proposition 4.1. It illustrates the case $p = 5$, $g_X = 5$, and $f_X = 4$. Each dot represents a one-dimensional subspace of $H^1_{dR}(Y)$ (the upper group) or $H^1_{dR}(X)$ (the lower group). On $H^1_{dR}(Y)$, the action of δ shifts a given one-dimensional subspace to the one represented by the dot below (if there is one, otherwise to zero). The groups on the left represent the multiplicative parts, those on the right represent the étale parts, and those in the middle represent the local-local parts. The (canonical) class $\eta_X \in H^1_{dR}(X)_{\acute{e}t}$ is constructed in the proof and spans the image of the injection in part (1) (i.e., the subspace $H^1_{dR}(X)^1$). The (non-canonical) class $\omega_X \in H^1_{dR}(X)_m$ maps onto a class spanning the image of the projection in part (1) (i.e., the subquotient $H^1_{dR}(X)/H^1_{dR}(X)^2$). We may choose ω_X so that $\langle \omega_X, \eta_X \rangle = 1$. The (non-canonical) classes ω_Y and η_Y can be chosen to satisfy $\pi^*\omega_X = \omega_Y$, $\pi_*\eta_Y = \eta_X$, and $\langle \omega_Y, \eta_Y \rangle = 1$. Caution: In general, the lines spanned by ω_X, ω_Y, and η_Y are not invariant under $\mathbb{D}_k$. This is closely related to the possible non-splitting of the exact sequences (1.5) and (1.6).

Proof of Proposition 4.1. (1) Since $\pi : Y \to X$ is unramified, the choice of a fixed isomorphism $\operatorname{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ makes Y into a $\mathbb{Z}/p\mathbb{Z}$-torsor over X. The group that classifies such torsors is $H^1_{\acute{e}t}(X, \mathbb{Z}/p\mathbb{Z})$ (see, for example, [Mil25, III.4] or [Poo17, 6.5.5]). Therefore, there is a class in $H^1_{\acute{e}t}(X, \mathbb{Z}/p\mathbb{Z})$ defined by the cover $\pi : Y \to X$ and the fixed isomorphism $G \cong \mathbb{Z}/p\mathbb{Z}$, which will be denoted by $\eta_{X,\acute{e}t}$.

Consider the exact sequence

$$0 \to \mathbb{Z}/p\mathbb{Z} \to \mathcal{O}_X \xrightarrow{\wp} \mathcal{O}_X \to 0$$

of sheaves for the étale topology on X where $\wp(x) = x^p - x$. Taking cohomology yields a homomorphism

$$H^1_{\acute{e}t}(X, \mathbb{Z}/p\mathbb{Z}) \to H^1_{\acute{e}t}(X, \mathcal{O}_X)[\wp],$$

where $[\wp]$ indicates the kernel of $\wp$. Since $H^0(Y, \mathcal{O}_Y) = k$, the image of $\eta_{X,\acute{e}t}$ in $H^1_{\acute{e}t}(X, \mathcal{O}_X)[\wp]$ is non-zero, and since $H^1_{\acute{e}t}(X, \mathcal{O}_X)[\wp]$ is the subset of the usual (coherent) $H^1(X, \mathcal{O}_X)$ fixed by Frobenius, we have a class $\eta_{X,coh} \in H^1(X, \mathcal{O}_X)$ fixed by Frobenius. It has a canonical lift to $H^1_{dR}(X)$ (take any lift and apply F), and we denote this lift by η_X. Since $F\eta_X = \eta_X$, we see that $\eta_X \in H^1_{dR}(X)_{\acute{e}t}$, and the injection in (1) is $\alpha \mapsto \alpha\eta_X$.

The surjection in (1) is obtained from the injection by Cartier duality, and is given more explicitly by $c \mapsto \langle c, \eta_X \rangle_X$.

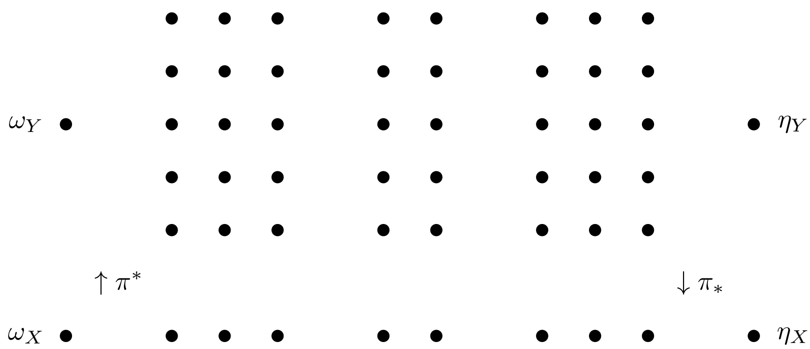

FIGURE 1. de Rham cohomology as G-module

Decomposing the module $\mathcal{M}_X$ into its étale, multiplicative, and local-local parts shows that

$$\mathcal{M}_{X,ét} = \mathrm{Coker}\left(M(\mathbb{Z}/p\mathbb{Z}) \to H^1_{dR}(X)_{ét}\right),$$
$$\mathcal{M}_{X,m} = \mathrm{Ker}\left(H^1_{dR}(X)_m \to M(\mu_p)\right),$$

and

$$\mathcal{M}_{X,ll} = H^1_{dR}(X)_{ll}.$$

Thus $\mathcal{M}_X$ is the Dieudonné module of $\mathcal{G}_X$. This establishes part (1) of the Proposition.

(2) Taking the multiplicative and étale parts of the de Rham sequence (4.1) with $Z = Y$ yields isomorphisms

$$H^1_{dR}(Y)_m \cong H^0(Y, \Omega^1_Y)_m \quad \text{and} \quad H^1_{dR}(Y)_{ét} \cong H^1(Y, \mathcal{O}_Y)_{ét}.$$

Tamagawa [Tam51] proves that there is an isomorphism of $k[G]$-modules

$$H^0(Y, \Omega^1_Y) \cong V_1 \oplus V_p^{g_X - 1}, \tag{4.4}$$

and Serre duality yields

$$H^1(Y, \mathcal{O}_Y) \cong V_1 \oplus V_p^{g_X - 1}.$$

Nakajima [Nak85] proves that there is an isomorphism of $k[G]$-modules

$$H^1_{dR}(Y)_m \cong H^0(Y, \Omega^1_Y)_m \cong V_1 \oplus V_p^{f_X - 1}, \tag{4.5}$$

and Cartier duality yields

$$H^1_{dR}(Y)_{ét} \cong H^1(Y, \Omega^1_Y)_{ét} \cong V_1 \oplus V_p^{f_X - 1}.$$

This establishes the first two claims in part (2).

For the third claim in part (2), let $h_X = g_X - f_X$, and note that the last four displayed equations and (4.1) show that $H^1_{dR}(Y)_{ll}$ is an extension of $V_p^{h_X}$ by $V_p^{h_X}$. Since V_p is free, the extension splits and there is an isomorphism of $k[G]$-modules

$$H^1_{dR}(Y)_{ll} \cong V_p^{2h_X}.$$

The fourth claim in part (2) is simply the direct sum of the three preceding claims. This completes the proof of part (2) of the proposition.

(3) Consider the Hochschild–Serre spectral sequence for π in de Rham cohomology. The sequence of low degree terms is

$$0 \to H^1(G, H^0_{dR}(Y)) \to H^1_{dR}(X) \overset{\pi^*}{\to} H^1_{dR}(Y)[\delta] \to H^2(G, H^0_{dR}(Y)).$$

We have

$$H^1(G, H^0_{dR}(Y)) \cong H^2(G, H^0_{dR}(Y)) \cong M(\mathbb{Z}/p\mathbb{Z}),$$

and these modules are étale, so taking multiplicative and local-local parts of the sequence yields isomorphisms

$$H^1_{dR}(X)_m \overset{\pi^*}{\to} H^1_{dR}(Y)_m[\delta] \quad \text{and} \quad H^1_{dR}(X)_{ll} \overset{\pi^*}{\to} H^1_{dR}(Y)_{ll}[\delta].$$

Taking the étale part yields an exact sequence

$$0 \to M(\mathbb{Z}/p\mathbb{Z}) \to H^1_{dR}(X)_{ét} \overset{\pi^*}{\to} H^1_{dR}(Y)_{ét}[\delta] \to M(\mathbb{Z}/p\mathbb{Z}) \to 0, \tag{4.6}$$

where surjectivity on the right follows from a dimension count using part (2). Thus η_X spans the kernel of π^* in (4.6). We will prove the last claim in (3) after establishing (4).

(4) The Cartier dual of the isomorphism

$$H^1_{dR}(X)_m \overset{\pi^*}{\to} H^1_{dR}(Y)_m[\delta]$$

is the isomorphism

$$H^1_{dR}(Y)_m/\delta H^1_{dR}(Y)_m \overset{\pi_*}{\to} H^1_{dR}(X)_m.$$

It follows from part (2) that the image of the natural map

$$H^1_{dR}(Y)_{ét}[\delta] \to H^1_{dR}(Y)_{ét}/\delta H^1_{dR}(Y)_{ét}$$

is a line, and that it is equal to

$$\mathrm{Ker}\left(H^1_{dR}(Y)_{ét}/\delta H^1_{dR}(Y)_{ét} \overset{\delta^{p-1}}{\to} H^1_{dR}(Y)_{ét}[\delta]\right),$$

and therefore equal to

$$\mathrm{Ker}\left(H^1_{dR}(Y)_{ét}/\delta H^1_{dR}(Y)_{ét} \overset{\pi^*\pi_*}{\to} H^1_{dR}(Y)_{ét}[\delta]\right).$$

This shows that π_* identifies this line with

$$\mathrm{Ker}\left(H^1_{dR}(X)_{ét} \overset{\pi^*}{\to} H^1_{dR}(Y)_{ét}[\delta]\right),$$

and we observed above that this is the line spanned by η_X, i.e., the line defined in part (1). This completes the proof of part (4). We also deduce that

$$\pi^*\left(H^1_{dR}(X)_{ét}\right) = \pi^*\pi_*\left(H^1_{dR}(Y)_{ét}\right) = \delta^{p-1}\left(H^1_{dR}(Y)_{ét}\right),$$

and this establishes the last claim in part (3). □

We now turn to an elegant result that holds under the assumption that X has a k-rational point.

Proposition 4.3. *Assume that X has a k-rational point S and let $T = \pi^{-1}(S)$. Let $\mathcal{N}_Y$ be the hypercohomology group*

$$\mathcal{N}_Y := \mathbb{H}^1\left(Y, \mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)\right).$$

Then

(1) *$\mathcal{N}_Y \cong V_p^{2g_X}$ as $k[G]$-modules.*
(2) *For $i = 1, \ldots, p$, there are isomorphisms*

$$\frac{\mathcal{N}_Y[\delta^i]}{\mathcal{N}_Y[\delta^{i-1}]} = \frac{\delta^{p-i}\mathcal{N}_Y}{\delta^{p-i+1}\mathcal{N}_Y} \cong H^1_{dR}(X)$$

of $\mathbb{D}_k$-modules.
(3) *There are exact sequences of $\mathbb{D}_k[G]$-modules*

$$0 \to H^0(Y, \mathcal{O}_Y) \to H^0(Y, \mathcal{O}_T) \to \mathcal{N}_{Y,ét} \to H^1_{dR}(Y)_{ét} \to 0,$$

and

$$0 \to H^1_{dR}(Y)_m \to \mathcal{N}_{Y,m} \to H^0(Y, \Omega^1_Y(T)/\Omega^1_Y) \to H^1(Y, \Omega^1_Y) \to 0,$$

as well as an isomorphism

$$\mathcal{N}_{Y,ll} \cong H^1_{dR}(Y)_{ll}.$$

The $\mathbb{D}_k[G]$-module structures and homomorphisms in part (3) will be made explicit in the proof.

Remark 4.4.

(1) In parallel with Remark 4.2, each of the subquotients $\mathcal{N}_Y[\delta^i]/\mathcal{N}_Y[\delta^j]$ for $0 \le i < j \le p$, is self-dual, but not in a way compatible with restrictions of pairings.
(2) The exact sequences appearing in part (3) may also be interpreted as a 3-step filtration on $\mathcal{N}_Y$ with

$$\begin{aligned}\mathcal{N}_Y^3 &= \mathcal{N}_Y,\\ \mathcal{N}_Y^2 &= \ker\left(\mathcal{N}_Y \to H^0(Y, \Omega^1_Y(T)/\Omega^1_Y)\right),\\ \mathcal{N}_Y^1 &= \operatorname{Im}\left(H^0(Y, \mathcal{O}_T) \to \mathcal{N}_Y\right),\end{aligned}$$

and

$$\mathcal{N}_Y^0 = 0.$$

The subquotients are

$$\operatorname{Ker}\left(H^0(Y, \Omega^1_Y(T)/\Omega^1_Y) \to H^1(Y, \Omega^1_Y)\right),$$
$$H^1_{dR}(Y), \quad \text{and} \quad \operatorname{Coker}\left(H^0(Y, \mathcal{O}_Y) \to H^0(Y, \mathcal{O}_T)\right).$$

(3) The filtration $\mathcal{N}_Y^i$ is self-dual in the sense that $\mathcal{N}_Y^2$ and $\mathcal{N}_Y^1$ are orthogonal complements to one another.
(4) The filtration on $\mathcal{N}_Y$ induces a filtration on each of the subquotients $\mathcal{N}_Y[\delta^i]/\mathcal{N}_Y[\delta^{i+1}]$, and the induced filtration is the one on $H^1_{dR}(X)$ in Remark 4.2.

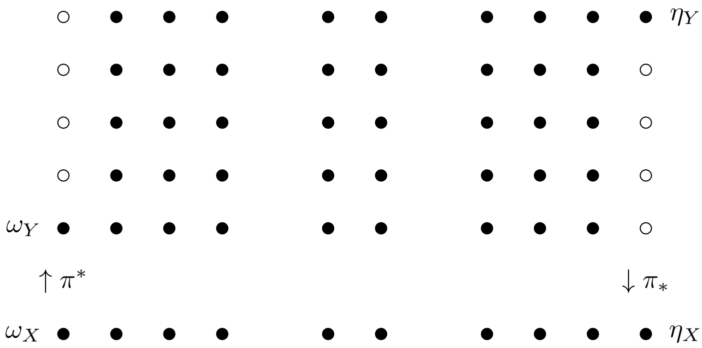

FIGURE 2. Modified de Rham cohomology as G-module

Figure 2 illustrates the case $g_X = 5$, $f_X = 4$, and $p = 5$, with the same conventions as in Figure 1. In this case, $\pi_* : \mathcal{N}_Y/\delta\mathcal{N}_Y \to H^1_{dR}(X)$ is an isomorphism, as is $\pi^* : H^1_{dR}(X) \to \mathcal{N}_Y[\delta]$. The open circles on the right represent the submodule $\mathcal{N}_Y^1$ and the open circles on the left represent the quotient module $\mathcal{N}_Y^3/\mathcal{N}_Y^2$. Note that the classes ω_X, η_Y, and ω_Y are not canonically defined.

Proof of Proposition 4.3. (1) By [Nak85, Thm. 1], $H^0(Y, \Omega^1_Y(T))$ is free over $k[G]$ of rank g_X. The same is true of $H^1(Y, \mathcal{O}_Y(-T))$ by Serre duality. The modified de Rham exact sequence

$$0 \to H^0(Y, \Omega^1_Y(T)) \to \mathbb{H}^1\left(Y, \mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)\right) \to H^1(Y, \mathcal{O}_Y(-T)) \to 0$$

shows that $\mathcal{N}_Y = \mathbb{H}^1\left(Y, \mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)\right)$ is $k[G]$-free of rank $2g_X$.

(2) The quotients appearing in the statement are all isomorphic (via a suitable power of δ) to $\mathcal{N}_Y[\delta]$, so it will suffice to prove that $\mathcal{N}_Y[\delta] \cong H^1_{dR}(X)$ as $\mathbb{D}_k$-modules.

Note that $\pi^*\mathcal{O}_X(-S) = \mathcal{O}_Y(-T)$ and $\pi^*\Omega^1_X(S) = \Omega^1_Y(T)$. The exact sequence of low degree terms of the Hochschild-Serre spectral sequence for $Y \to X$ and $\mathcal{O}_X(-S)$ yields an isomorphism

$$\pi^* : H^1(X, \mathcal{O}_X(-S)) \xrightarrow{\sim} H^1(Y, \mathcal{O}_Y(-T))[\delta],$$

and since $Y \to X$ is unramified, it is clear that we have an isomorphism

$$\pi^* : H^0(X, \Omega^1_X(S)) \xrightarrow{\sim} H^0(Y, \Omega^1_Y(T))[\delta].$$

Using the modified de Rham sequences of $\mathcal{O}_X(-S) \xrightarrow{d} \Omega^1_X(S)$ and $\mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)$ shows that π^* induces an isomorphism

$$\pi^* : \mathbb{H}^1\left(X, \mathcal{O}_X(-S) \xrightarrow{d} \Omega^1_X(S)\right) \xrightarrow{\sim} \mathbb{H}^1\left(Y, \mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)\right)[\delta].$$

Since $\deg S = 1$, we also have $H^1(X, \mathcal{O}_X(-S)) \cong H^1(X, \mathcal{O}_X)$, and $H^0(X, \Omega^1_X(S)) \cong H^0(X, \Omega^1_X)$, so

$$H^1_{dR}(X) = \mathbb{H}^1\left(\mathcal{O}_X \xrightarrow{d} \Omega^1_X\right) \cong \mathbb{H}^1\left(X, \mathcal{O}_X(-S) \xrightarrow{d} \Omega^1_X(S)\right).$$

This completes the proof of part (2).

(3) Note that there are exact sequences of complexes of coherent sheaves on Y:

$$0 \to \left(\mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y\right) \to \left(\mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y(T)\right) \to \left(\Omega^1_Y(T)/\Omega^1_Y\right)[-1] \to 0,$$

and

$$0 \to \left(\mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y\right) \to \left(\mathcal{O}_Y \xrightarrow{d} \Omega^1_Y\right) \to \mathcal{O}_T \to 0.$$

Let $\mathcal{N}^2_Y := \mathbb{H}^1\left(Y, \mathcal{O}_Y(-T) \xrightarrow{d} \Omega^1_Y\right)$ and $\mathcal{N}^1_Y := \operatorname{Coker}\left(H^0(Y, \mathcal{O}_Y) \to H^0(T, \mathcal{O}_T)\right)$.

Taking cohomology of the first exact sequence above, we have

$$0 \to \mathcal{N}^2_Y \to \mathcal{N}_Y \to H^0(Y, \Omega^1_Y(T)/\Omega^1_Y) \to H^1(Y, \Omega^1_Y) \to 0, \tag{4.7}$$

where the surjection on the right is the sum of residues map. Note that $H^0(Y, \Omega^1_Y(T)/\Omega^1_Y)$ is $k[G]$-free of rank 1, and $H^1(Y, \Omega^1_Y)$ is k with the trivial $k[G]$ action. The action of F on both of these spaces is zero, and the V action is induced by the Cartier operator on differentials.

Taking cohomology of the second exact sequence above, we have

$$0 \to H^0(Y, \mathcal{O}_Y) \to H^0(Y, \mathcal{O}_T) \to \mathcal{N}^2_Y \to H^1_{dR}(Y) \to 0. \tag{4.8}$$

Note that $H^0(Y, \mathcal{O}_T)$ is $k[G]$-free of rank 1, and $H^0(Y, \mathcal{O}_Y)$ is k with the trivial $k[G]$ action. The action of F on both of these spaces is the usual action of Frobenius, and the V action is trivial.

The last two displayed exact sequences give the asserted filtration on $\mathcal{N}_Y$, and this completes the proof of part (3) of the proposition. □

Remark 4.5.

(1) The hypercohomology group $\mathcal{N}_Y$ is closely related to the de Rham cohomology of the singular curve associated to Y and T where T is viewed as a "modulus", as in [Ser88, Ch. 4].

(2) Suppose that $g_X > 1$ and choose a k-rational, effective, reduced divisor S on X. A canonical divisor of X is k-rational and effective, and since k is perfect, the underlying reduced divisor of a k-rational divisor is again k-rational, so there always exists a divisor S as above with degree $\leq 2g_X - 2$. Let $T = \pi^{-1}(S)$. Then the proof of Proposition 4.3 applies essentially verbatim and shows that $\mathcal{N}_Y$ is free over $k[G]$ and that

$$\mathcal{N}_Y[\delta] \cong \mathcal{N}_X := \mathbb{H}^1\left(X, \mathcal{O}_X(-S) \xrightarrow{d} \Omega^1_X(S)\right).$$

Thus, at the expense of enlarging $H^1_{dR}(X)$ and $H^1_{dR}(Y)$ by certain simple étale and multiplicative $\mathbb{D}_k$-modules, we can always arrange that the cohomology associated to Y is $k[G]$-free with subquotients isomorphic to the cohomology associated to X.

(3) We may also choose two reduced, strictly effective divisors S_1 and S_2 on X, set $T_i = \pi^*(S_i)$ and take hypercohomology of the complexes

$$\mathcal{O}_X(-S_1) \to \Omega^1_X(S_2) \quad \text{and} \quad \mathcal{O}_Y(-T_1) \to \Omega^1_Y(T_2).$$

Then the latter is a free $k[G]$-module with subquotients isomorphic to the former.

5. Proofs of Proposition 1.1 and Theorems 1.2 and 1.5

Proof of Proposition 1.1. The group scheme homomorphisms in the first sentence of Proposition 1.1 are obtained by applying the Dieudonné functor to the $\mathbb{D}_k$-module homomorphisms in part (1) of Proposition 4.1. This establishes the claims in Proposition 1.1, and it identifies $\mathcal{M}_X$ as the Dieudonné module of $\mathcal{G}_X$. $\square$

Proof of Theorem 1.2. (1) Splitting the 4-term exact sequence in part (3) of Proposition 4.1 into two parts and using the identification of the image of π_* there, we obtain

$$0 \to M(\mathbb{Z}/p\mathbb{Z}) \to H^1_{dR}(X)_{\acute{e}t} \to \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t} \to 0,$$

and

$$0 \to \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t} \to H^1_{dR}(Y)_{\acute{e}t}[\delta] \to M(\mathbb{Z}/p\mathbb{Z}) \to 0.$$

The first of these yields an isomorphism $\mathcal{M}_{X,\acute{e}t} \cong \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t}$. Using part (4) of Proposition 4.1, we obtain the diagram

$$\begin{tikzcd}
0 \arrow[r] & M(\mathbb{Z}/p\mathbb{Z}) \arrow[r] & H^1_{dR}(X)_{\acute{e}t} \arrow[r, "\pi^*"] & \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t} \arrow[r] & 0 \\
0 \arrow[r] & M(\mathbb{Z}/p\mathbb{Z}) \arrow[r] \arrow[u, equal] & H^1_{dR}(Y)_{\acute{e}t}/\delta \arrow[r, "\delta^{p-1}"] \arrow[u, "\pi_*"'] & \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t} \arrow[r] \arrow[u, equal] & 0.
\end{tikzcd}$$

Using the isomorphism $\mathcal{M}_{X,\acute{e}t} \cong \delta^{p-1}H^1_{dR}(Y)_{\acute{e}t}$ and applying the Diedonné functor yields the exact sequence (1.5) of Theorem 1.2 and an identification of it with the exact sequence (1.3). Similarly, the second exact sequence above yields (1.6).

Part (2) of Proposition 4.1 shows that the subquotients

$$\frac{\delta^i H^1_{dR}(Y)_{\acute{e}t}}{\delta^{i+1}H^1_{dR}(Y)_{\acute{e}t}} \quad \text{for } i = 1\dots,p-1 \quad \text{and} \quad \frac{H^1_{dR}(Y)_{\acute{e}t}[\delta^i]}{H^1_{dR}(Y)_{\acute{e}t}[\delta^{i-1}]} \quad \text{for } i = 2\dots,p$$

are all isomorphic to one another via a suitable power of δ. Since $\delta^{p-1}H^1_{dR}(Y)_{\acute{e}t} \cong \mathcal{M}_{X,\acute{e}t}$, applying the Dieudonné functor yields the isomorphisms in the first sentence of part (1) of Theorem 1.2, and this completes the proof of this part.

Part (2) is equivalent to part (1) by Cartier duality.

For part (3), note that part (2) of Proposition 4.1 shows that the subquotients

$$\frac{\delta^i H^1_{dR}(Y)_{ll}}{\delta^{i+1}H^1_{dR}(Y)_{ll}} \quad \text{for } i = 1\dots,p \quad \text{and} \quad \frac{H^1_{dR}(Y)_{ll}[\delta^i]}{H^1_{dR}(Y)_{ll}[\delta^{i-1}]} \quad \text{for } i = 1\dots,p$$

are all isomorphic to one another via a suitable power of δ. Moreover, parts (1) and (3) of Proposition 4.1 show that

$$\mathcal{M}_{X,ll} \cong H^1_{dR}(X)_{ll} \cong H^1_{dR}(Y)_{ll}[\delta],$$

so all of the subquotients above are isomorphic to $\mathcal{M}_{X,ll}$. Applying the Dieudonné functor yields part (3) of Theorem 1.2. This completes the proof of that theorem. $\square$

Proof of Theorem 1.5. Let $\mathcal{H}$ be the k group scheme with Dieudonné module $\mathcal{N}_Y$ as defined in Proposition 4.3. By part (1) of that proposition, $\mathcal{H}$ is G-free, and we have equalities $\delta^i\mathcal{H} = \mathcal{H}[\delta^{p-i}]$ for $i = 1, \ldots, p$. By part (2), there are canonical isomorphisms

$$\frac{\delta^i\mathcal{H}}{\delta^{i+1}\mathcal{H}} \cong \frac{\mathcal{H}[\delta^{p-i}]}{\mathcal{H}[\delta^{p-i-1}]} \cong J_X[p] \quad \text{for } i = 0, \ldots, p-1.$$

Note that

$$M\left(\mathrm{Res}_{T/S}\,\mathbb{Z}/p\mathbb{Z}\right) \cong H^0(Y, \mathcal{O}_T),$$

where the right hand side is a k vector space on which F acts by the p-power Frobenius and $V = 0$, and that

$$M\left(\mathrm{Res}_{T/S}\,\mu_p\right) \cong H^0\left(Y, \Omega^1_Y(T)/\Omega^1_Y\right),$$

where the right hand side is a k vector space on which $F = 0$ and V acts by the Cartier operator (which is essentially the inverse Frobenius on residues). Then applying the Dieudonné functor to part (3) of Proposition 4.3 yields the exact sequences asserted in part (3) of Theorem 1.5. This completes the proof of the theorem. $\square$

6. Comments on $\mathbb{D}_k[G]$-modules and examples

6.1. **Motivation.** Suppose that N is a finite-dimensional k-vector space equipped with an action of $\mathbb{D}_k$ and/or $G = \mathbb{Z}/p\mathbb{Z}$. Then N is both Artinian and Noetherian, so the Krull-Schmidt theorem holds: N is the direct sum of indecomposable submodules, and the number and isomorphism types of the summands are uniquely determined. (See, for example, [Jac89, §3.4].)

In the case where $N = H^1_{dR}(Y)$ or $\mathcal{N}_Y$, we have complete information on N as a $k[G]$-module from part (2) of Proposition 4.1 or part (1) of Proposition 4.3 respectively. If we take $H^1_{dR}(X)$ as given, then we know the associated graded of N with respect to the two filtrations attached to the G-action by the proofs of Theorems 1.2 and 1.5 in Section 5.

The basic question that motivates Sections 7 and 8 is this: what restrictions on N as a $\mathbb{D}_k$-module are imposed by the information in the preceding paragraph? As we will see below, this question appears to be quite difficult, and we are only able to give satisfactory answers in a few cases. In the rest of this section, we explain some of the difficulties and give examples illustrating them.

6.2. **$H^1_{dR}(Y)$ is not determined by $H^1_{dR}(X)$.** We show by example that $H^1_{dR}(Y)$ is not determined by $H^1_{dR}(X)$, even when k is algebraically closed; that is, $H^1_{dR}(Y)$ is not determined as a $\mathbb{D}_k[G]$-module by its associated graded $\mathbb{D}_k$-module.

Assume that k is algebraically closed. Then, as explained in the proof of part (1) of Proposition 4.1, the data of the cover $\pi : Y \to X$ and the isomorphism $\mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ determines and is determined by an element in the finite-dimensional $\mathbb{F}_p$-vector spaces

$$H^1_{et}(X, \mathbb{Z}/p\mathbb{Z}) \cong H^1_{dR}(X)^{F=1} \cong H^1(X, \mathcal{O}_X)^{F=1}.$$

Multiplying the element by a scalar in $\alpha \in {\mathbb{F}_p}^\times$ represents the same cover $\pi : Y \to X$ with the isomorphism $\mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ multiplied by α. The set of unramified covers $\pi : Y \to X$ (without the isomorphism $\mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$) is thus in bijection with the projective space $\mathbb{P}\left(H^1(X, \mathcal{O}_X)^{F=1}\right)$. For general k, elements of $\mathbb{P}\left(H^1(X,\mathcal{O}_X)^{F=1}\right)$ correspond to $\overline{k}$-isomorphism classes of covers which can be defined over k. (In general, they are represented by several distinct k-isomorphism classes of covers.)

Take $p=3$ and let X be the degree 9 hyperelliptic curve over $k:=\mathbb{F}_3$ given by

$$X: \qquad y^2 = x^9 + x^4 + x^2 + 1.$$

Then X has genus $g_X = 4$ and $f_X = \nu_X = 2$, and $H^1(X,\mathcal{O}_X)^{F=1}$ is 2-dimensional over $\mathbb{F}_p$. The 4 unramified covers of X over $\overline{k}$ thus all arise from covers defined over k. For each, we choose a cover representing it given by $Y_i:\ z^p - z = f_i$, with f_i in the table below, and for each cover Y_i we record the EO-type of $\mathbb{D}(J_{Y_i}[p]_{ll})$ (which determines the isomorphism class of $J_{Y_i}[p]_{ll}$ over $\overline{k}$, but not in general over k). For $a \in \mathbb{F}_3$, we also consider the twisted curve $Y_i(a): z^p - z = f_i + a$ and record the invariant factors of F acting on $\mathbb{D}(J_{Y_i}(a)[p]_{et})$, which determine its isomorphism type as a $k[F]$-module. We emphasize that $Y_i(a) \simeq Y_i$ over $\overline{k}$ for each a. This list of invariant factors will be of the form $(F-1)^{e_1}, (F-1)^{e_2}, \ldots$, with $e_1 \le e_2 \le \ldots$, and for ease of notation we record it simply as $e_1, e_2, \ldots$. Perhaps surprisingly, the isomorphism type varies among the three possible twists over k.

$Y_i:\ z^p - z =$	EO-type of $\mathbb{D}(J_{Y_i}[p]_{ll})$	Inv. factors of F on $\mathbb{D}(J_{Y_i(a)}[p]_{ét})$		
		$a=0$	$a=1$	$a=-1$
$-(x^6+x^3+x-1)y$	$[0,0,0,1,2,3]$	$1,1,2$	$1,3$	$1,3$
$(x^6+x+1)y$	$[0,0,1,1,2,3]$	$1,1,2$	$1,3$	$1,3$
$(x^3+1)y+1$	$[0,1,1,2,3,4]$	$1,3$	$1,3$	$1,1,2$
$(x^6-x^3+x)y$	$[0,1,1,2,3,4]$	$1,1,2$	$1,3$	$1,3$

Note that the three group schemes $J_{Y_i}[p]_{ll}$ for $1 \le i \le 3$ are pairwise non-isomorphic over $\overline{k}$, and that the four k-group schemes $J_{Y_i}[p]$ for $1 \le i \le 4$ are pairwise non-isomorphic over k. However, $J_{Y_3}[p]$ and $J_{Y_4}[p]$ become isomorphic over $\overline{k}$. Note as well that the EO-types of $J_{Y_3}[p]$ and $J_{Y_4}[p]$ show that the conclusion of Theorem 1.10 need not hold when $g_X - f_X > 1$.

6.3. **Extensions of BT_1 modules.** Propositions 4.1 and 4.3 tell us that the self-dual BT_1 module $H^1_{dR}(Y)_{ll}$ is a repeated extension of the self-dual BT_1 module $H^1_{dR}(X)_{ll}$. In this section, we make some comments (surely well-known to experts) about how ill-behaved such extensions may be, even when k is algebraically closed.

More precisely, consider the full subcategory of the category of $\mathbb{D}_k$-modules whose objects are BT_1 modules. We use freely the Kraft classification of BT_1 modules in terms of cyclic words on the two letter alphabet $\{f,v\}$. (See [PU21] for an overview.)

It is a simple exercise to check that this category is closed under extension and quotient in the sense that if

$$0 \to M_1 \to M \to M_2 \to 0$$

is an exact sequence of $\mathbb{D}_k$-modules, and if two of M, M_1, M_2 are BT_1 modules, then so is the third.

However, this category has the unfortunate property that the image and kernel of a morphism between BT_1 modules need not be BT_1 modules. For example, if $M_{1,1}$ is the module associated to the word fv (so one generator e and one relation $Fe = Ve$), then the maps of modules $M \to M$ determined by sending e to Fe have kernel and image isomorphic to the module $\mathbb{D}_k/(F,V)$, and this is not a BT_1.

In [Oor05], there is a determination of the simple BT_1 modules, i.e., those that have no non-trivial BT_1 submodule. It follows by standard arguments that every BT_1 module is an iterated extension of simple BT_1 modules.

Unfortunately, there is no Jordan-Hölder theorem here: The list of simple BT_1 modules appearing in a presentation of a given M as an extension of simple BT_1 modules is not in general uniquely determined.

Here is an example: Let M be the module associated to f^3v^3 (one generator e with relation $F^3e = V^3e$). Then it is not hard to check that M is a three-fold extension of $M_{1,1}$. On the other hand, M also admits a surjection onto the module $M_{2,1}$ corresponding to ffv (one generator a with relation $F^2a = Va$) with kernel isomorphic to $M_{1,2}$, the module corresponding to the word fvv (one generator b with relation $Fb = V^2b$). Oort's results imply that $M_{2,1}$ and $M_{1,2}$ are simple, so we have no uniqueness of "Jordan-Hölder" factors.

6.4. $\mathbb{D}_k[G]$**-modules.** We show by example that there may be infinitely many non-isomorphic $\mathbb{D}_k[G]$-modules with given $\mathbb{D}_k$- and $k[G]$-module structures. We first show that (for any p), the $\mathbb{D}_k$-module associated to the word f^2v^2 has infinitely many non-isomorphic presentations as an extension of the module associated to fv by itself. Recall [PU21, §3.2] that $M(fv)$ is given by one generator e_1 and one relation $Fe_1 = Ve_1$, and that $M(f^2v^2)$ is given by one generator e_2 and one relation $F^2e_2 = V^2e_2$. A homomorphism $\phi : M(f^2v^2) \to M(fv)$ is determined by its value on e_2 which is in turn determined by two arbitrary elements of k:

$$\phi(e_2) = \alpha e_1 + \beta F e_1,$$

and the surjective homomorphisms are those with $\alpha \neq 0$. The kernel of a surjective ϕ is the subspace spanned by $\alpha^{1/p}Fe_2 - \alpha^p Ve_2$ and F^2e_2. One checks that this subspace is a $\mathbb{D}_k$-module isomorphic to $M(fv)$. Note that the kernels of distinct ϕ are in general distinct subspaces of $M(f^2v^2)$; in fact, they are in general not even conjugate by $\mathrm{Aut}_{\mathbb{D}_k}(M(f^2v^2)) \cong \mathbb{F}_{p^4} \times k^3$.

Now assume that $p = 2$. For each choice of surjective ϕ as above, we may equip $M = M(f^2v^2)$ with the structure of a free $R = k[G]$ module of rank 2 in such a way that $M[\delta] = \ker\phi$. (See Lemma 8.12.) This shows that there are infinitely many non-isomorphic $\mathbb{D}_k[G]$-modules with the same underlying R-module (namely R^2) and the same underlying $\mathbb{D}_k$ module (namely $M(f^2v^2)$).

7. Applications to the étale part of $J_Y[p]$

We turn to a consideration of the étale part of $J_Y[p]$. When k is algebraically closed, $J_Y[p]_{ét}$ is completely determined by the isomorphism (1.1). As we will see below, the situation is more interesting when k is only assumed to be perfect.

Proof of Proposition 1.6. Applying the Dieudonné functor to parts (3) and (4) of Proposition 4.1 yields an exact sequence

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{ét}/\delta \xrightarrow{\delta^{p-1}} J_Y[p]_{ét}[\delta] \to \mathbb{Z}/p\mathbb{Z} \to 0,$$

and an identification of $\mathrm{Im}\,\delta^{p-1}$ with $\mathcal{G}_X$ via the isomorphism $\pi^* : J_X[p]_{ét} \to J_Y[p]_{ét}[\delta]$. Splitting this exact sequence into two short exact sequences yields sequences (1.5) and (1.6).

For part (1), assume that the sequence (1.5) splits. Then we have an inclusion

$$\mathbb{Z}/p\mathbb{Z} \hookrightarrow J_Y[p]_{ét}[\delta] \hookrightarrow J_Y[p]_{ét}$$

whose image does not lie in $\operatorname{Im}\delta^{p-1}$. By part (2) of Proposition 4.1, the image therefore does not lie in the image of δ. This shows that the quotient $\mathcal{Q}$ defined by the exactness of

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{ét} \to \mathcal{Q} \to 0 \tag{7.1}$$

is G-free.

Conversely, if we have the exact sequence (7.1) with $\mathcal{Q}$ assumed to be G-free, then the image of $\mathbb{Z}/p\mathbb{Z} \to J_Y[p]$ is killed by δ and not in the image of δ, so it splits (1.5). This completes the proof of part (1) of the proposition.

For part (2), assume that the sequence (1.6) splits. Then we have a surjection

$$J_Y[p]_{ét} \twoheadrightarrow J_Y[p]_{ét}/\delta \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}$$

whose kernel maps surjectively to $\mathcal{G}_X$ via δ^{p-1}. This shows that the subgroup $\mathcal{K}$ defined by the exactness of

$$0 \to \mathcal{K} \to J_Y[p]_{ét} \to \mathbb{Z}/p\mathbb{Z} \to 0 \tag{7.2}$$

is G-free.

Conversely, if we have the exact sequence (7.2) with $\mathcal{K}$ assumed to be G-free, then the surjection $J_Y[p]_{ét} \to \mathbb{Z}/p\mathbb{Z}$ factors through $J_Y[p]_{ét}/\delta$, so it splits (1.6). This completes the proof of part (2) of the proposition.

For part (3), if the exact sequences (1.5) and (1.6) both split, then we have the sequences (7.1) and (7.2). It then follows easily from part (2) of Theorem 4.1 that the composed maps

$$\mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{ét} \to \mathbb{Z}/p\mathbb{Z}$$

and

$$\mathcal{K} \to J_Y[p]_{ét} \to \mathcal{Q}$$

are isomorphisms, so $J_Y[p]_{ét}$ is isomorphic to the direct sum of $\mathbb{Z}/p\mathbb{Z}$ and a G-free group. (Moreover, sequences (7.1) and (7.2) both split.) The converse is straightforward.

This completes the proof of Proposition 1.6. □

Example 7.1. In general, the splittings of exact sequences (1.5) and (1.6) appear to be independent conditions: Table 1 provides several examples of étale $\mathbb{Z}/3\mathbb{Z}$-covers of genus 3 hyperelliptic curves over $\mathbb{F}_3$ with $f_X = 2$ exhibiting that all 4 splitting possibilities indeed occur. (The additional notations d_1, d_2, and μ are explained in Example 7.5.) Note, however, that in certain special situations, there are implications: for example when k is finite and $\nu_X = 1$, the splitting of (1.5) implies that of (1.6) by Theorem 1.7(4).

In some of the proofs below, it will be convenient to use the language of Galois representations. Recall (e.g., using [Poo17, §5.8]) that there is an equivalence of categories between étale p-group schemes over k and representations of $\operatorname{Gal}(\overline{k}/k)$ on finite-dimensional $\mathbb{F}_p$-vector spaces. The representation associated to a group $\mathcal{G}$ is the $\mathbb{F}_p$-vector space $\mathcal{V} := \mathcal{G}(\overline{k})$ equipped with the natural action of $\operatorname{Gal}(\overline{k}/k)$. In particular, when k is finite, $\operatorname{Gal}(\overline{k}/k)$ is pro-cyclic, and $\mathcal{G}$ is determined by a single endomorphism of $\mathcal{V}$, namely Frobenius.

TABLE 1. $p = 3,\ f_X = 2$

$X : y^2 =$	ν_X	$Y : z^3 - z =$	ν_Y	(1.5) split?	(1.6) split?	d_1	d_2	μ
$x^7 + x^5 + x$	1	$(x^4 + x^2 + x)y$	1	yes	yes	2	1	3
$-x^7 - x^3 + x$	1	xy	1	yes	yes	2	3	2
$-x^7 - x^3 - 1$	1	$xy - 1$	1	yes	yes	2	3	1
$x^7 + x^6 + x^4 + x$	1	$(x + 1)y$	2	no	no	3	1	
$x^7 + x^6 + x^4 + 1$	1	$(x + 1)y + 1$	2	no	yes	3	1	
$-x^7 + x^6 + x^4 + x$	1	$(-x + 1)y$	3	no	yes	3	1	
$x^8 - x^6 + x^4 - x^2 + 1$	2	$(x^2 - 1)y - x^6 + x^2 - 1$	2	yes	yes	1	3	
$-x^7 + x^6 + x^2 + x$	2	$(-x + 1)y$	3	yes	no	1	3	
$-x^7 + x^6 + x^2 + x$	2	$(-x^4 + x^3 + x - 1)y$	3	yes	yes	1	3	
$-x^7 + x^6 + x^2 + x$	2	$(-x^4 + x^3)y$	4	yes	yes	1	1	

Proof of Theorem 1.7. First note that if

$$0 \to \mathcal{G}_1 \to \mathcal{G}_2 \to \mathcal{G}_3 \to 0$$

is an exact sequence of p-torsion group schemes over k, then

$$\nu(\mathcal{G}_1) \le \nu(\mathcal{G}_2) \le \nu(\mathcal{G}_1) + \nu(\mathcal{G}_3).$$

If the sequence splits, then $\nu(G_2) = \nu(\mathcal{G}_1) + \nu(\mathcal{G}_3)$.

By part (1) of Theorem 1.2,

$$J_X[p]_{\acute{e}t} \cong J_Y[p]_{\acute{e}t}[\delta] \subset J_Y[p]_{\acute{e}t},$$

so we have $\nu_X \le \nu_Y$.

Again by part (1) of Theorem 1.2, we have exact sequences

$$0 \to J_Y[p]_{\acute{e}t}[\delta^{i-1}] \to J_Y[p]_{\acute{e}t}[\delta^i] \to \mathcal{G}_X \to 0$$

for $i = 2, \ldots, p$.

Applying the observation in the first paragraph of the proof and induction on i, we find that $\nu_Y \le \nu_X + (p-1)\nu(\mathcal{G}_X)$. Since $\mathcal{G}_X$ is a subgroup scheme of $J_X[p]$, $\nu(\mathcal{G}_X) \le \nu_X$ and we have $\nu_Y \le p\nu_X$. This establishes part (1).

If (1.3) splits, then $\nu(\mathcal{G}_X) = \nu_X - 1$ and we find that $\nu_Y \le p(\nu_X - 1) + 1$. This yields part (2).

For part (3), if k is algebraically closed, then $\nu_X = f_X$ and $f_X \ge 1$ since we have assumed X has an étale $\mathbb{Z}/p\mathbb{Z}$ cover. Now assume that k is finite. Note that by Proposition 1.1, the representation associated to $J_X[p]_{\acute{e}t}$ has the trivial representation as a quotient. This is equivalent to saying that the action of Frobenius on $\mathcal{V} = J_X[p](\overline{k})$ has 1 as an eigenvalue. It follows that $J_X[p](k)$ is non-trivial, i.e., $\nu_X \ge 1$.

If k is finite, (1.5) is split, and (1.6) is non-split, then we claim that $\mathrm{Fr}_k - 1$ cannot act bijectively on $\mathcal{G}_{X,\acute{e}t}(\overline{k})$, where Fr_k generates $\mathrm{Gal}(\overline{k}/k)$. Indeed, assuming to the contrary that $\mathrm{Fr}_k - 1$ is bijective on $\mathcal{G}_{X,\acute{e}t}(\overline{k})$, one sees via the snake lemma that in the exact sequence of $\overline{k}$-points associated to (1.6), the image of $\mathrm{Fr}_k - 1$ on $(J_Y[p]_{\acute{e}t}/\delta)(\overline{k})$ projects isomorphically onto the quotient $\mathcal{G}_{X,\acute{e}t}(\overline{k})$; the inverse of this isomorphism splits (1.6), contradicting our hypothesis. We conclude that Fr_k must have a non-trivial fixed vector on $\mathcal{G}_{X,\acute{e}t}(\overline{k})$. Since (1.3) (equivalently (1.5)) splits by hypothesis, we deduce that the space of Fr_k-fixed vectors in $J_X[p]_{\acute{e}t}(\overline{k})$ is at least 2-dimensional, *i.e.* $\nu_X \ge 2$. This completes the proof of part (4) of the theorem. ∎

Example 7.2. Table 1 shows that the bounds on ν_Y in Theorem 1.7(1)–(2) are sharp: if (1.5) is non-split, we must have $\nu_X = 1$ since $f_X = 2$ throughout the table (indeed, the alternative is $\nu_X = 2 = f_X$, in which case $J_X[p]_{ét}$ would be completely split, implying the splitting of (1.5)). This forces $\nu_Y > 1$ by Theorem 1.9(B) and Proposition 1.6(3), which gives the bounds $1 = \nu_X < \nu_Y \leq 3$ in this situation. Lines 4–6 of the table show that both possibilities $\nu_Y = 2, 3$ indeed occur.

When (1.5) is split, we have $\nu_X \leq \nu_Y \leq 3\nu_X - 2$; lines 1–3 show that the unique possibility $\nu_Y = 1$ indeed occurs when $\nu_X = 1$, while lines 7–10 show that all three possibilities $\nu_Y = 2, 3, 4$ occur when $\nu_X = 2$.

Example 7.3. In part (3), if we do not assume that k is finite, $\mathrm{Gal}(\overline{k}/k)$ may no longer be procyclic, and it may have $\mathbb{F}_p$-representations with trivial quotients, but no trivial subrepresentations. Here is an example of such a representation.

Take $p = 3$, let $F = \mathbb{F}_p(t)$, let $F' = \mathbb{F}_p(v)$, and embed $F \hookrightarrow F'$ by $t \mapsto (v^3 - v)^2$. Then one checks that F'/F is Galois with group S_3. Let k and k' be the perfections of F and F' respectively. Then we have

$$\mathrm{Gal}(\overline{k}/k) \twoheadrightarrow \mathrm{Gal}(k'/k) \cong S_3.$$

Let γ_1 and γ_2 be two involutions generating S_3, and let S_3 act on $\mathcal{V} := \mathbb{F}_3^2$ by the matrices

$$\gamma_1 \mapsto \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \gamma_2 \mapsto \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then one sees easily that the resulting representation of $\mathrm{Gal}(\overline{k}/k)$ has the trivial representation as a quotient, but it does not have the trivial representation as a sub. Thus the corresponding group scheme $\mathcal{G}$ admits a surjection to $\mathbb{Z}/p\mathbb{Z}$, but has $\nu(\mathcal{G}) = 0$.

To prove Theorems 1.8 and 1.9, we will use an approach via coordinates, so we begin with some preliminaries on $\mathbb{F}_p[G]$-modules.

Let $S = \mathbb{F}_p[G] \cong \mathbb{F}_p[\delta]/(\delta^p)$, and for $i = 1, \ldots, p$, let $W_i = S/(\delta^i)$ be the i-dimensional indecomposable module over S. Write $a \mapsto \overline{a}$ for the natural reduction homomorphism $S \to k$ (the quotient modulo δ). We have natural identifications $\mathrm{Hom}_S(W_p, W_p) \cong S$, $\mathrm{Hom}_S(W_p, W_1) \cong \mathbb{F}_p$, $\mathrm{Hom}_S(W_1, W_p) \cong \mathbb{F}_p$, and $\mathrm{Hom}_S(W_1, W_1) \cong \mathbb{F}_p$. Under these identifications, the maps

$$\mathrm{Hom}_S(W_p, W_p) \to \mathrm{Hom}_S(W_p[\delta], W_p[\delta]) = \mathrm{Hom}_S(W_1, W_1)$$

and

$$\mathrm{Hom}_S(W_p, W_p) \to \mathrm{Hom}_S(W_p/\delta, W_p/\delta) = \mathrm{Hom}_S(W_1, W_1)$$

are both the reduction map $S \to k$. The restriction map

$$\mathrm{Hom}_S(W_p, W_1) \to \mathrm{Hom}_S(W_p[\delta], W_1) = \mathrm{Hom}_S(W_1, W_1)$$

is zero, as is the reduction map

$$\mathrm{Hom}_S(W_1, W_p) \to \mathrm{Hom}_S(W_1, W_p/\delta) = \mathrm{Hom}_S(W_1, W_1).$$

Now let $\mathcal{V}_X$ and $\mathcal{V}_Y$ be the representations of $\mathrm{Gal}(\overline{k}/k)$ associated to $J_X[p]_{ét}$ and $J_Y[p]_{ét}$:

$$\mathcal{V}_X = J_X[p]_{ét}(\overline{k}) \quad \text{and} \quad \mathcal{V}_Y = J_Y[p]_{ét}(\overline{k}).$$

By part (2) of Proposition 4.1, we have a (non-canonical) isomorphism of S-modules

$$\mathcal{V}_Y \cong W_1 \oplus W_p^{f_X - 1}.$$

Choosing such an isomorphism, we may represent the action of $\phi \in \mathrm{Gal}(\overline{k}/k)$ by a matrix of the form

$$\left(\begin{array}{c|c} a_0 & \alpha_1 \cdots \alpha_r \\ \hline \beta_1 & \\ \vdots & A = (a_{ij}) \\ \beta_r & \end{array}\right), \tag{7.3}$$

where $r = f_X - 1$, $a_0 \in \mathrm{Hom}_S(W_1, W_1)$, $\alpha_j \in \mathrm{Hom}_S(W_p, W_1)$, $\beta_i \in \mathrm{Hom}_S(W_1, W_p)$, $a_{ij} \in \mathrm{Hom}_S(W_p, W_p)$, and i and j run from 1 to r.

Using the observations above on S-homomorphisms, we find that the induced action of ϕ on $\mathcal{V}_Y[\delta] \cong \mathcal{V}_X$ is given by the matrix

$$\left(\begin{array}{c|c} a_0 & 0 \cdots 0 \\ \hline \beta_1 & \\ \vdots & \overline{A} = (\overline{a}_{ij}) \\ \beta_r & \end{array}\right), \tag{7.4}$$

and the induced action of ϕ on $\mathcal{V}_Y/\delta$ is given by the matrix

$$\left(\begin{array}{c|c} a_0 & \alpha_1 \cdots \alpha_r \\ \hline 0 & \\ \vdots & \overline{A} = (\overline{a}_{ij}) \\ 0 & \end{array}\right).$$

The block triangular structure of the last two displayed matrices reflects the exact sequences (1.5) and (1.6) (i.e., the fact that $\mathbb{Z}/p\mathbb{Z}$ is a quotient of $\mathcal{V}_Y[\delta]$ and a sub of $\mathcal{V}_Y/\delta$), and we find that $a_0 = 1$. Moreover, (1.5) splits if and only if we may choose the isomorphism $\mathcal{V}_Y \cong W_1 \oplus W_p^r$ such that the β_i all vanish, and (1.6) splits if and only if we may choose the isomorphism such that the α_j all vanish. (This gives an alternate proof of Proposition 1.6.)

Lemma 7.4. *Define U, the 1-units of $\mathrm{GL}_r(S)$, by the exactness of*

$$0 \to U \to \mathrm{GL}_r(S) \to \mathrm{GL}_r(\mathbb{F}_p) \to 0,$$

where the surjection is $(a_{ij}) \mapsto (\overline{a}_{ij})$. Then the group U has exponent p.

Proof. Write an element of U in the form $I + \Delta M$ where Δ is the diagonal matrix with all diagonal entries equal to δ and where $M \in M_r(S)$. Since I and Δ are in the center of the characteristic p ring $M_r(S)$, we have

$$(I + \Delta M)^p = I + \Delta^p M^p = I.$$

□

With these preliminaries, we are ready to prove the two remaining theorems about $J_Y[p]_{ét}$.

Proof of Theorem 1.8. For part (1), note that if ϕ has matrix of the shape (7.4) with $a_0 = 1$, then by an inductive argument, ϕ^n has matrix

$$\left(\begin{array}{c|c} 1 & 0 \cdots 0 \\ \hline n\beta_1 & \\ \vdots & \overline{A}^n \\ n\beta_r & \end{array}\right).$$

Thus there is a finite Galois extension k' of k with $\mathrm{Gal}(k'/k)$ of exponent p such that the representation of $\mathrm{Gal}(\overline{k}/k')$ on $\mathcal{V}_X = \mathcal{V}_Y[\delta]$ has the trivial representation as a direct factor, i.e., such that the sequences (1.3) and (1.5) split over k'.

For part (2), to say that $J_X[p]_{ét}$ is completely split is to say that for every $\phi \in \mathrm{Gal}(\overline{k}/k)$, the matrix of ϕ is of the form (7.3) with $a_0 = 1$, $\beta_i = 0$ for all i, and $\overline{A} = I$, i.e., with $A \in U$. Lemma 7.4 then shows that ϕ^p acts trivially on $\mathcal{V}_Y$. This shows that there is a finite Galois extension k' of k with $\mathrm{Gal}(k'/k)$ of exponent p such that $J_Y[p]_{ét}$ is completely split over k'.

This completes the proof of the theorem. $\square$

Example 7.5. Table 1 (in which each base curve X has $f_X = 2$) illustrates that the degree bounds of Theorems 1.8 and 1.9 are sharp: in the antepenultimate column we have listed the degree d_1 of the unique minimal extension k_1/k with the property that $J_X[p]_{ét}$ is completely split over k_1. The penultimate column lists the degree d_2 of the minimal extension k_2/k_1 over which $J_Y[p]_{ét}$ is completely split. When $\nu_X = \nu_Y = 1$, the final column gives the positive integer μ specified in (1a) of Theorem 1.9; note that all possibilities indeed occur. The first three lines of the table provides examples in which the extension k' guaranteed by (1a) of Theorem 1.9 is as large as possible; in the first line $k'' = k$, while in the second and third lines k'' is the unique degree $p = 3$ extension of k.

Proof of Theorem 1.9. For part (A), if $f_X = 1$, then in equation (1.3), $\mathcal{G}_X = 0$ and $J_X[p]_{ét} \cong \mathbb{Z}/p\mathbb{Z}$. It then follows from part (1) of Theorem 1.2 that $J_Y[p]_{ét} \cong \mathbb{Z}/p\mathbb{Z}$.

Now consider parts (B) and (C). The alternatives are mutually exclusive, and they exhaust the possibilities, so it will suffice to verify the additional claims in each case. Note that the subgroup $\mathcal{G}_{X,ét}$ defined by exact sequence (1.3) corresponds to a line in $\mathcal{V}_X$ which is invariant under $\mathrm{Gal}(\overline{k}/k)$. Let $\phi \in \mathrm{Gal}(\overline{k}/k)$ be Frobenius, and let $\rho \in \mathbb{F}_p{}^\times$ be the eigenvalue of ϕ on this line.

(Case (1a)) If $\rho \neq 1$, then $p > 2$ since $\mathbb{F}_2^\times = \{1\}$. Moreover, both (1.5) and (1.6) are split (by taking the kernel or image of a high power of $\phi - \rho$). Thus, we may choose the isomorphism $\mathcal{V}_Y \cong W_1 \oplus W_p$ so that ϕ has the shape

$$\left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & a \end{array}\right),$$

where $a \in S$ has $\overline{a} = \rho$. It is then clear that $\nu_X = \nu_Y = 1$ and we are in case (1a). The group scheme $\mathcal{Q}$ in the statement is the one corresponding to the representation of $\mathrm{Gal}(\overline{k}/k)$ on W_p with Frobenius acting by a. Since $\overline{a}^{p-1} = \rho^{p-1} = 1$, over an extension k' of k of degree dividing $p-1$, ϕ acts on W_p via a 1-unit, so has a non-trivial space of invariants. This shows that $|\mathcal{Q}(k')| = p^\mu$ with $1 \leq \mu \leq p$. Since a is ρ times a 1-unit, Lemma 7.4 shows that $a^p = \rho^p = \rho \in k \subset S$, so that over the extension k'' of degree p, ρ acts on W_p by ρ, and $\mathcal{G} \cong (\mathcal{G}')^p$ for a rank 1, non-split group scheme $\mathcal{G}'$. Finally, $\phi^{p(p-1)}$ acts trivially on $\mathcal{V}_Y$, so $J_Y[p]_{ét}$ is completely split over an extension of degree dividing $p(p-1)$.

(Case (1b), $p > 2$, and Case (1), $p = 2$) Next, suppose that $\rho = 1$ and (1.3) is not split. Then $\mathcal{G}_X \cong \mathbb{Z}/p\mathbb{Z}$ and $\nu_X = 1$. If $p = 2$, this is enough to conclude that we are in case (1). Since $J_X[p]_{ét}$ is not completely split, $J_Y[p]_{ét}$ is also not completely split and $\nu_Y < p+1$. The action of ϕ is given by a matrix of the form

$$\left(\begin{array}{c|c} 1 & \alpha \\ \hline \beta & a \end{array}\right),$$

where $\beta \neq 0$. If $p > 2$, we consider the matrix of ϕ with respect to a suitable k-basis of $W_1 \oplus W_p$, namely 1 for W_1 and $\delta^{p-1}, \delta^{p-2}, \dots, 1$ for W_p. Then ϕ takes the form

$$\left(\begin{array}{c|ccccc} 1 & 0 & \dots & 0 & \alpha \\ \hline \beta & 1 & * & * & * \\ 0 & 0 & 1 & * & * \\ \vdots & \vdots & \ddots & 1 & * \\ 0 & 0 & \dots & 0 & 1 \end{array}\right).$$

It is then visible that $\phi - 1$ has rank $< p$, so $\nu_Y > 1$. Thus we are in case (1b). For any p, an inductive argument shows that ϕ^n has matrix

$$\left(\begin{array}{c|c} 1 & n\alpha \\ \hline n\beta & (1 + 2 + \cdots + (n-1))\beta\alpha + a^n \end{array}\right).$$

Applying Lemma 7.4 shows that if $p > 2$, then $J_Y[p]_{\acute{e}t}$ is completely split over the extension of k degree p, and if $p = 2$, then $J_Y[p]_{\acute{e}t}$ is completely split over the extension of k degree dividing 4.

(Case (2)) Finally, if $\rho = 1$ and (1.3) splits, then $\nu_X = 2$, $J_X[p]_{\acute{e}t}$ splits completely, and we are in case (2) (for any p). The conclusions there follow from Theorem 1.7 and part (2) of Theorem 1.8. We can say a bit more about the structure of $J_Y[p]_{\acute{e}t}$ in this case: By part (1) of Proposition 1.6, there is an exact sequence

$$0 \to \mathbb{Z}/p\mathbb{Z} \to J_Y[p]_{\acute{e}t} \to \mathcal{Q} \to 0,$$

where $\mathcal{Q}$ is G-free of rank 1. We have $\nu(\mathcal{Q}) \geq 1$, and both the extension above and the group scheme $\mathcal{Q}$ split completely over an extension of degree dividing p.

This completes the proof of Theorem 1.9. □

Example 7.6. Taking into account Theorem 1.7(2), which implies that when k is finite and $\nu_X = 1$, the splitting of (1.5) implies that of (1.6), we see that Table 1 exhibits that all possibilities specified by Theorem 1.9 indeed occur when $p = 3$ and $k = \mathbb{F}_3$.

7.1. **Dependence of $\mathcal{H}$ and $\mathcal{N}_Y$ on S.** The group scheme $\mathcal{H}$ in Theorem 1.5 and its Dieudonné module $\mathcal{N}_Y$ (analyzed in Proposition 4.3) apparently depend on the choice of a rational point S. Indeed, the subquotients $\mathrm{Ker}\left(\mathrm{Res}_{T/S}\,\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}\right)$ and $\mathrm{Coker}\left(\mu_p \to \mathrm{Res}_{T/S}\,\mu_p\right)$ of $\mathcal{H}$ and their Dieudonné modules

$$\mathrm{Coker}\left(k = H^0(Y, \mathcal{O}_Y) \to H^0(Y, \mathcal{O}_T)\right) \quad \text{and}$$
$$\mathrm{Ker}\left(H^0(Y, \Omega^1_Y(T)/\Omega^1_Y) \to H^1(Y, \Omega^1_Y) = k\right)$$

visibly depend on whether S splits in Y (and more precisely on the class of $T = \pi^{-1}(S)$ in $H^1(S, \mathbb{Z}/p\mathbb{Z})$).

Proposition 7.7. *If k is algebraically closed, then the isomorphism class of $\mathcal{N}_Y$ as a $\mathbb{D}_k[G]$-module is independent of the choice of the rational point S.*

Proof. The exact sequences (4.7) and (4.8) show that the local-local part of $\mathcal{N}_Y$ is independent of S (without any hypothesis on k). If k is algebraically closed, the étale part of $\mathcal{N}_Y$ is completely split as a $\mathbb{D}_k$-module (isomorphic to $M(\mathbb{Z}/p\mathbb{Z})^{pf_X}$) and by Proposition 4.3, it is G-free of rank f_X, so it is isomorphic to

$$M(\mathbb{Z}/p\mathbb{Z})^{f_X} \otimes \mathbb{F}_p[G],$$

and is thus independent of S. The same follows for the multiplicative part by Cartier duality. Since $\mathcal{N}_Y$ is the direct sum of its étale, multiplicative, and local-local parts, this establishes the proposition. $\square$

Example 7.8. Surprisingly, when k is finite, $\mathcal{N}_Y$ depends on S, even when S splits in Y. To emphasize the subtlety of this dependence on S and its fiber $T := \pi^{-1}(S)$ in Y, let us write $\mathcal{N}_Y(T)$ in place of $\mathcal{N}_Y$. Consider the smooth projective genus 5 hyperelliptic curve over $k = \mathbb{F}_3$ given by the affine equation

$$X: \quad y^2 + x^{12} + x^{10} - x^9 + x^6 + x^4 - x^2 + x - 1 = 0.$$

This curve has a-number 1 and arithmetic p-rank $\nu_X = 2$, so in particular has two independent unramified $\mathbb{Z}/3\mathbb{Z}$-covers. One such cover Y is given by the Artin–Schreier equation $z^3 - z = f$ where f is the rational function

$$\begin{aligned} f = &\frac{x^9 - x^7 - x^6 + x^5 + x^4 + x^3 + x^2 - x + 1}{x^{15}} y \\ &- \frac{x^{15} + x^{14} - x^{13} + x^{12} + x^{11} - x^{10} + x^9 + x^6 - x^3 + 1}{x^{15}}. \end{aligned}$$

The projective curve X has exactly six k-rational points (x, y):

$$S_1 := (0,-1), S_2 := (0,1), S_3 := (-1,-1), S_4 := (-1,1), S_5 := (1,-1), S_6 := (1,1);$$

note that the point at infinity on X is of degree 2. Let $\pi : Y \to X$ be the covering map and $T_i := \pi^{-1} S_i$ be the fiber in Y over S_i. Each T_i gives a $\mathbb{Z}/3\mathbb{Z}$-torsor (for the étale topology) over k, so gives a class in $H^1_{ét}(\mathrm{Spec}(k), \mathbb{Z}/p\mathbb{Z})$. Via the canonical identifications of abelian groups

$$H^1_{ét}(\mathrm{Spec}(k), \mathbb{Z}/p\mathbb{Z}) \simeq H^1(\mathrm{Gal}(\overline{k}/k), \mathbb{Z}/p\mathbb{Z}) \simeq k/\wp k = \mathbb{Z}/3\mathbb{Z},$$

each torsor T_i gives a class $[T_i] \in \mathbb{Z}/3\mathbb{Z}$. For example, $[T_1] = 0 = [T_3]$ as each of T_1 and T_3 consist of 3 distinct k-rational points of Y, whereas $[T_i] = \pm 1$ for $i = 2, 4, 5, 6$ since for these values of i, the fiber T_i is a single degree 3 point on Y.

Using MAGMA, we compute the matrix of V acting on the spaces $H^0(\Omega^1_Y(T_i))$ for $i = 1, \ldots, 6$. We also compute the action of the Artin–Schreier automorphism of $Y \to X$ on the residue field of T_i, which determines $[T_i]$, and obtain the following table:

i	1	2	3	4	5	6
$[T_i]$	0	1	0	1	-1	-1
$\dim\ker(V-1)$	4	3	4	3	3	3
$\dim\ker(V-1)^3$	9	9	8	8	8	8

It follows from this that the four $k[V]$-modules $H^0(\Omega^1_Y(T_i)) = \mathcal{N}_Y(T_i)[F]$ for $1 \le i \le 4$ are pairwise non-isomorphic. For $i = 1, 3$ we have short exact sequences of $k[V]$-modules

$$0 \longrightarrow H^0(Y, \Omega^1_Y) \longrightarrow H^0(Y, \Omega^1_Y(T_i)) \longrightarrow \left(\frac{k[V]}{(V-1)}\right)^2 \longrightarrow 0$$

while for $i = 2, 4$ we have short exact sequences

$$0 \longrightarrow H^0(Y, \Omega^1_Y) \longrightarrow H^0(Y, \Omega^1_Y(T_i)) \longrightarrow \frac{k[V]}{(V-1)^2} \longrightarrow 0.$$

Noting that the kernel of $V-1$ on $H^0(\Omega^1_Y)$ has dimension 3, our computations show that the above exact sequences are not only *non*-split for $1 \leq i \leq 4$, but that the two *extension* classes of $k[V]$-modules provided by $H^0(\Omega^1_Y(T_i))$ for $i = 1, 3$ (respectively $i = 2, 4$) are non-isomorphic! This is rather surprising, as all four $k[V]$-modules $H^0(\Omega^1_Y(T_i))$ for $1 \leq i \leq 4$ become isomorphic after a finite extension of the ground field. We conclude that the $\mathbb{D}_k$ modules $\mathcal{N}_Y(T_i)$ are non-isomorphic for $1 \leq i \leq 4$. On the other hand, further computation shows that $\mathcal{N}_Y(T_4) \simeq \mathcal{N}_Y(T_5) \simeq \mathcal{N}_Y(T_6)$ as $\mathbb{D}_k$-modules, which is again somewhat surprising as the torsors T_4 and T_5 are non-isomorphic, while the torsors T_1, T_3 *are* isomorphic, as are T_2, T_4. Again, all six Dieudonné modules $\mathcal{N}_Y(T_i)$ become isomorphic after a suitable finite extension on k.

8. Applications to $J_Y[p]$: The local-local part

In this section, we consider $J_Y[p]_{ll}$ and its Dieudonné module $H^1_{dR}(Y)_{ll}$. Throughout, we assume $k = \overline{k}$. We view the local-local part of $H^1_{dR}(X)$ as given, and we exploit the G-module structure on $H^1_{dR}(Y)_{ll}$ to find restrictions on its structure as a $\mathbb{D}_k$-module. We begin by recording the basic properties of $H^1_{dR}(Y)_{ll}$.

Proposition 8.1. *Write M for $H^1_{dR}(Y)_{ll}$ and let $h = g_X - f_X$. We have*

$$M[\delta] \cong M/\delta \cong H^1_{dR}(X)_{ll}.$$

Moreover, M has the following properties:

(1) *M is a free $k[G]$-module of rank $2h$.*
(2) *M is a self-dual, local-local BT_1 module, i.e.,* $\operatorname{Im} F = \operatorname{Ker} V$, $\operatorname{Im} V = \operatorname{Ker} F$, *$F$ and V act nilpotently on M, and M admits a perfect k-bilinear pairing $\langle \cdot, \cdot \rangle$ which is alternating ($\langle m, m \rangle = 0$ for all $m \in M$) and which satisfies $\langle Fm, n \rangle = \langle m, Vn \rangle^p$ for all $m, n \in M$.*
(3) *The pairing is compatible with the G action in that $\langle \gamma m, \gamma n \rangle = \langle m, n \rangle$ for all $m, n \in M$.*
(4) $\operatorname{Im} F = \operatorname{Ker} V$ *and* $\operatorname{Im} V = \operatorname{Ker} F$ *are free submodules of M of rank h.*

Proof. Proposition 4.1 shows that $M[\delta] \cong M/\delta \cong H^1_{dR}(X)_{ll}$. Part (1) was proven in Proposition 4.1, part (2). For part (2), as we reviewed at the beginning of Section 4, $H^1_{dR}(Y)$ is a self-dual BT_1 module where the duality is induced by the de Rham pairing, and it is easy to see that the restriction of this pairing makes M into a self-dual BT_1 module. It is local-local by definition. Part (3) holds because γ is an automorphism of Y, so has degree 1. For part (4), note that $VM = H^0(Y, \Omega^1_Y)_{ll}$, and comparing the result of Tamagawa (equation (4.4)) to that of Nakajima (equation (4.5)) shows that VM is free over $k[G]$ of rank h. The same then follows for $\operatorname{Im} F = \operatorname{Ker} V$ since $VM \cong M/(\operatorname{Ker} V)$. □

Remark 8.2. For a module M with properties (1)–(3), the spaces $\operatorname{Im} F = \operatorname{Ker} V$ and $\operatorname{Im} V = \operatorname{Ker} F$ are all free over $k[G]$ of rank h as soon as one of them is. Furthermore, in this situation the calculations

$$\langle Fm, Fn \rangle = \langle m, VFn \rangle^p = 0 \quad \text{and} \quad \langle Vm, Vn \rangle = \langle m, FVn \rangle^{1/p} = 0$$

show that $\operatorname{Im} F$ and $\operatorname{Im} V$ are isotropic, and they are maximal isotropic since they have dimension $ph = \frac{1}{2} \dim M$.

8.1. **Coordinates.** We will introduce special coordinates on any $\mathbb{D}_k[G]$-module M with properties (1)–(4) as in Proposition 8.1. This allows for numerical experimentation and leads to full analyses in certain significant cases.

Write R for the group ring $k[G]$ and introduce a k-linear involution $a \mapsto \tilde{a}$ by requiring that $\tilde{g} = g^{-1}$ for all $g \in G$. This involution is trivial if $p = 2$ and is non-trivial with invariant subspace of dimension $(p+1)/2$ if $p > 2$. We extend it to vectors and matrices with entries in R by acting component-wise.

Lemma 8.3. *If $p > 2$ and $a \in R$ is a non-zero element with $\tilde{a} = a$, then $a = \delta^\alpha u$ where $\alpha \in \{0, 2, \ldots, p-1\}$ (i.e., α is even) and $u \in R^\times$ (i.e., u is a unit).*

Proof. Since $\delta = \gamma - 1$, we have

$$\tilde{\delta} = \gamma^{-1} - 1 = -\gamma^{-1}\delta = -\delta/(1+\delta).$$

Writing $a = \delta^\alpha u$ where $\alpha \geq 0$ and $u \in R^\times$, we see that $\tilde{a} \equiv (-1)^\alpha a \pmod{\delta^{\alpha+1} R}$ so $\tilde{a} = a$ implies that α is even. □

Recall that $\gamma \in G$ is the element corresponding to 1 under $G \cong \mathbb{Z}/p\mathbb{Z}$. For $a = a_0 + a_1\gamma + \cdots + a_{p-1}\gamma^{p-1} \in R$, define

$$(a)_0 := a_0.$$

Next, note that the function $R \times R \to k$ given by

$$(a, b) := (a\tilde{b})_0$$

is k-bilinear and satisfies $(\gamma a, \gamma b) = (a, b)$. Let J be the $2h$-by-$2h$ matrix

$$J = \begin{pmatrix} 0_h & I_h \\ -I_h & 0_h \end{pmatrix},$$

where 0_h and I_h are the $h \times h$ zero and identity matrices respectively. Regarding R^{2h} as a space of column vectors, we have a perfect, alternating, k-bilinear pairing on R^{2h} given by

$$\langle m, n \rangle = \left({}^t m J \tilde{n}\right)_0.$$

Here, ${}^t m$ stands for the transpose of m and $\tilde{n}$ is computed by applying the involution $a \mapsto \tilde{a}$ to each entry of n. If G acts coordinate-wise on elements of R^{2h}, then $\langle \gamma m, \gamma n \rangle = \langle m, n \rangle$.

If M is a $\mathbb{D}_k[G]$-module which is free over R of rank $2h$, and if $m_1, \ldots, m_{2h}$ is an ordered basis of M, we write $[m]$ for the coordinate vector of m:

$$[m] = \begin{pmatrix} r_1 \\ \vdots \\ r_{2h} \end{pmatrix} \qquad \text{if } m = r_1 m_1 + \cdots + r_{2h} m_{2h}.$$

Since F acts R-semi-linearly (i.e., $F(am) = a^{(p)} Fm$ for $a \in R$ and $m \in M$), we may represent F by a $2h$-by-$2h$ matrix $\mathcal{F}$ with entries in R, namely, the matrix such that

$$[Fm] = \mathcal{F}[m]^{(p)},$$

where the right hand side is the matrix product of $\mathcal{F}$ and $[m]^{(p)}$.

We say that a $\mathbb{D}_k$-module N of dimension $2h$ over k is *superspecial* if it is isomorphic to the Dieudonné module of $E[p]^h$, with E a supersingular elliptic curve. Three equivalent characterizations are: (i) $F^2 = V^2 = 0$ on N; (ii) in the Kraft–Oort classification, N corresponds to the word fv repeated h times; and (iii) In the Ekedahl–Oort classification, N corresponds to the elementary sequence $[0, 0, \ldots, 0]$.

Proposition 8.4. *Suppose that M is a $\mathbb{D}_k[G]$-module with properties (1)–(4) as in Proposition 8.1.*

(1) *There exists an ordered basis $m_1, \ldots, m_{2h}$ of M such that the pairing $\langle \cdot, \cdot \rangle$ is given by*

$$\langle m, n \rangle = \left({}^t[m] J \widetilde{[n]} \right)_0 ,$$

and such that the matrix of F has the form

$$\mathcal{F} = \begin{pmatrix} 0 & B \\ 0 & D \end{pmatrix},$$

where B and D are h-by-h matrices with entries in R satisfying ${}^t\tilde{D}B = {}^t\tilde{B}D$ and the columns of $\mathcal{F}$ generate a free R-module of rank h.

(2) *Conversely, any choice of B and D satisfying ${}^t\tilde{D}B = {}^t\tilde{B}D$ and such that the columns of $\mathcal{F} = \begin{pmatrix} 0 & B \\ 0 & D \end{pmatrix}$ generate a free R-module of rank h and $\mathcal{F}$ is p-nilpotent[2] defines the structure of $\mathbb{D}_k[G]$-module on R^{2h} which satisfies properties (1)–(4) of Proposition 8.1.*

(3) *If $M/\delta M$ is superspecial, then we may choose the basis so that the matrix of F has the form*

$$\mathcal{F} = \begin{pmatrix} 0 & I \\ 0 & D \end{pmatrix},$$

where δ divides D (i.e., δ divides every entry of D) and ${}^t\tilde{D} = D$.

(4) *Conversely, any choice of D which is divisible by δ and which satisfies ${}^t\tilde{D} = D$ defines the structure of $\mathbb{D}_k[G]$-module on R^{2h} which satisfies properties (1)–(4) of Proposition 8.1 and which is superspecial modulo δ.*

Remark 8.5.

(1) In parts (1) and (3), we do not claim that B and D are uniquely determined by M.

(2) In parts (2) and (4), the action of V is determined by that of F and the pairing by the requirement that $\langle Fm, n \rangle = \langle m, Vn \rangle^p$.

Proof. By hypothesis, $\operatorname{Ker} F$ is free of rank h over R and isotropic for the pairing. Choose an R-basis $m_1, \ldots, m_h$ for $\operatorname{Ker} F$. Since the pairing is non-degenerate, we may choose elements $n_1, \ldots, n_h$ of M such that for $0 \leq i < p$ and $1 \leq j, \ell \leq h$ we have

$$\langle \gamma^i m_j, n_\ell \rangle = \begin{cases} 1 & \text{if } i = 0 \text{ and } j = \ell, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\langle \gamma^i m_j, \gamma^{i'} n_\ell \rangle = \langle \gamma^{i-i'} m_j, n_\ell \rangle$, we find that $n_1, \ldots, n_h$ generate a free R-module of rank h which is complementary to $\operatorname{Ker} F$ and in k-duality with $\operatorname{Ker} F$ via the pairing. We then inductively modify the n_j by elements of $\operatorname{Ker} F$ to make their R-span isotropic. More precisely, set $m_{h+1} = n_1$, choose $m_{12} \in \operatorname{Ker} F$ such that $\langle m_{h+1}, m_{12} \rangle = \langle m_{h+1}, m_{12} \rangle$ and set $m_{h+2} = n_2 - m_{12}$, etc. Then for $1 \leq i, j \leq h$ we have

$$\langle m_i, m_j \rangle = \langle m_{h+i}, m_{h+j} \rangle = 0 \quad \text{and} \quad \langle m_i, m_{h+j} \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases}$$

[2] We say $\mathcal{F}$ is "p-nilpotent" if $\mathcal{F}\mathcal{F}^{(p)} \cdots \mathcal{F}^{(p^a)} = 0$ for some $a > 0$.

The pairing $\langle\cdot,\cdot\rangle$ then has the desired form with respect to the basis $m_1,\dots,m_{2h}$.

Since the first h basis elements span the kernel of F, the matrix of F has the form

$$\mathcal{F}=\begin{pmatrix}0&B\\0&D\end{pmatrix},$$

where B and D are h-by-h matrices with coordinates in R. Since $\operatorname{Im}F$ is R-free of rank h, the columns of $\mathcal{F}$ generate a free R-module of rank h. Let $\mathcal{V}$ be the matrix of V with respect to the chosen basis. The compatibility $\langle Fm,n\rangle=\langle m,Vn\rangle^p$ implies that ${}^t\mathcal{F}J=J\tilde{\mathcal{V}}^{(p)}$, so

$$\mathcal{V}=\begin{pmatrix}{}^t\tilde{D}^{(1/p)}&-{}^t\tilde{B}^{(1/p)}\\0&0\end{pmatrix},$$

and $VF=0$ implies $\mathcal{V}\mathcal{F}^{(1/p)}=0$ which in turn implies ${}^t\tilde{D}B={}^t\tilde{B}D$. This completes the proof of part (1).

For part (2), given B and D satisfying the conditions, define a p-linear operator F and a p^{-1}-linear operator $\mathcal{V}$ on R^{2h} by setting

$$Fm=\mathcal{F}m^{(p)}\quad\text{and}\quad Vm=\mathcal{V}m^{(1/p)},$$

where

$$\mathcal{F}=\begin{pmatrix}0&B\\0&D\end{pmatrix}\quad\text{and}\quad\mathcal{V}=\begin{pmatrix}{}^t\tilde{D}^{(1/p)}&-{}^t\tilde{B}^{(1/p)}\\0&0\end{pmatrix}.$$

Then $\operatorname{Im}F=\operatorname{Ker}V$ and $\operatorname{Im}V=\operatorname{Ker}F$, so we obtain a BT_1-module with a perfect alternating pairing, and it is straightforward to check that it has the properties (1)–(4) enumerated in Proposition 8.1. This completes the proof of part (2).

For part (3), first choose a basis as in part (1). Since $M/\delta M$ is superspecial, the matrix D is divisible by δ, and the condition that the columns of $\mathcal{F}$ span a free R-module of rank h implies that B is invertible.

Now consider changes of coordinates that preserve the matrix of the pairing. These are precisely the matrices $S\in GL_{2h}(R)$ satisfying ${}^tSJ\tilde{S}=J$. In particular, we may take S of the form

$$S=\begin{pmatrix}T&0\\0&U\end{pmatrix},$$

where $T,U\in\mathrm{GL}_h(R)$ and ${}^tT\tilde{U}=I$. In terms of the new basis, the matrix of F is

$$S^{-1}\mathcal{F}S^{(p)}=\begin{pmatrix}0&T^{-1}BU^{(p)}\\0&U^{-1}DU^{(p)}\end{pmatrix}=\begin{pmatrix}0&{}^t\tilde{U}BU^{(p)}\\0&U^{-1}DU^{(p)}\end{pmatrix}.$$

By the Lang–Steinberg theorem [Ste68, Thm. 10.1] (applied to the endomorphism $U\mapsto\left({}^t\tilde{U}^{-1}\right)^{(p)}$ of $\mathrm{GL}_h(R)$), we may choose U so that ${}^t\tilde{U}BU^{(p)}=I$. In these new coordinates, $\mathcal{F}$ has the desired form, and this proves part (3).

Part (4) follows from the same argument as in part (2) and the observation that $\mathcal{F}$ is p-nilpotent as soon as δ divides D. ☐

Remark 8.6. The coordinates in Proposition 8.4 provide another way to see that there can be infinitely many non-isomorphic $\mathbb{D}_k[G]$-modules with the same underlying $\mathbb{D}_k$-module and $k[G]$-module. In brief, one considers the subgroup of $\mathrm{GL}_{2h}(R)$ preserving the form and the subspace $\operatorname{Ker}F$. Assuming that $M/\delta M$ is superspecial and passing to the subgroup preserving the shape of $\mathcal{F}$ in part (3) (i.e., with an identity matrix in the upper right), we find matrices of block form $\begin{pmatrix}T&U\\0&W\end{pmatrix}$ where (among other conditions) ${}^t\tilde{W}W^{(p)}$ is congruent to the identity modulo δ. Such a matrix W is congruent modulo δ to one with entries in $\mathbb{F}_{p^2}$. On the other hand, the effect of this change of coordinates on D is $D \mapsto W^{-1}DW^{(p)}$, so the claim follows if one knows that the $\mathbb{D}_k$-module structure of M is determined by fairly crude invariants of D such as its rank and the rank of (Frobenius-twisted) iterates. We will not write down a general result of this shape, but the proofs of Theorem 8.7 and part (1) of Theorem 8.13 give examples where this is visible.

The coordinates of Proposition 8.4 can be used to analyze the special case where $h = 1$. Recall that the a-number of a $\mathbb{D}_k$-module is $a = \dim_k(\operatorname{Ker} F \cap \operatorname{Ker} V)$.

Theorem 8.7. *Assume $p > 2$ and let M be a $\mathbb{D}_k[G]$-module satisfying the properties (1)–(4) of Proposition 8.1 with $h = 1$. The a-number of M is in $\{2, 4, \ldots, p-1, p\}$. Define integers ℓ and b by $0 \le b < a$ and $p = \ell a + b$. Then*

$$M \cong M\left(f^{\ell+1}v^{\ell+1}\right)^b \bigoplus M\left(f^{\ell}v^{\ell}\right)^{a-b}.$$

In other words, M can be presented as the $\mathbb{D}_k$-module with generators $e_1, \ldots, e_a$ and relations

$$F^{\ell+1}e_i = V^{\ell+1}e_i \quad \text{for } 1 \le i \le b \quad \text{and} \quad F^{\ell}e_i = V^{\ell}e_i \quad \text{for } b < i \le a.$$

We give other descriptions of M in the proof below.

Proof. Since $M/\delta M$ is 2-dimensional over k and local-local, it is superspecial. We apply part (3) of Proposition 8.4 to identify M with R^2 where F acts via a matrix of the form $\begin{pmatrix} 0 & 1 \\ 0 & D \end{pmatrix}$ with $D \in R$ satisfying $\tilde{D} = D$, and V acts via $\begin{pmatrix} D^{(1/p)} & -1 \\ 0 & 0 \end{pmatrix}$.

We will use the Kraft–Oort and Ekedahl–Oort classifications of $\mathbb{D}_k$-modules to analyze M. (See [Oor01] for the original analysis and [PU21] for a more leisurely and detailed exposition.) First, we construct the canonical filtration on R^2, and then we check that under the Ekedahl–Oort classification, M has elementary sequence $[0, \ldots, 0, 1, 2, \ldots, p-a]$ (of length p with a zeroes), or equivalently under the Kraft–Oort classification, it corresponds to the words $f^{\ell+1}v^{\ell+1}$ with multiplicity b and $f^{\ell}V^{\ell}$ with multiplicity $a - b$.

We first treat the edge case $D = 0$. If $D = 0$, then one easily checks that the canonical filtration on M has the form

$$0 = M_0 \subset M_1 = R\begin{pmatrix} 1 \\ 0 \end{pmatrix} \subset M_2 = M,$$

with $FM = M_1$ and $FM_1 = 0$. Thus M has elementary sequence $[0, \ldots, 0]$, and a number p. In the Kraft–Oort classification, this corresponds to the word fv with multiplicity p.

Now assume that $D \neq 0$. Then D has the form $\delta^a u$ where $0 < a < p$ and $u \in R^\times$. Moreover, by Lemma 8.3, a must be even. Writing $p = \ell a + b$ with $b < a$, we have that $b \neq 0$. Define submodules $M_j \subset M = R^2$ for $0 \leq j \leq 4\ell + 2$ by

$$\begin{aligned}
M_{2i} &= R\delta^{p-ia}\begin{pmatrix}1\\D\end{pmatrix} && \text{for } 0 \leq i \leq \ell,\\
M_{1+2i} &= R\delta^{p-b-ia}\begin{pmatrix}1\\D\end{pmatrix} && \text{for } 0 \leq i \leq \ell,\\
M_{2\ell+1+2i} &= R\begin{pmatrix}1\\D\end{pmatrix} + R\delta^{(p-ia)}\begin{pmatrix}0\\1\end{pmatrix} && \text{for } 0 \leq i \leq \ell,\\
M_{2\ell+2+2i} &= R\begin{pmatrix}1\\D\end{pmatrix} + R\delta^{(p-b-ia)}\begin{pmatrix}0\\1\end{pmatrix} && \text{for } 0 \leq i \leq \ell.
\end{aligned}$$

Then we have

$$0 = M_0 \subset M_1 \subset \cdots \subset M_{2\ell+1} = FM \subset M_{2\ell+2} \subset \cdots \subset M_{4\ell+2} = M.$$

Next, one checks that

$$FM_j = \begin{cases} M_0 & \text{if } 0 \leq j \leq 2,\\ M_{j-2} & \text{if } 2 \leq j \leq 2\ell+1,\\ M_{2\ell-1} & \text{if } 2\ell+1 \leq j \leq 4\ell,\\ M_{2\ell} & \text{if } j = 4\ell+1,\\ M_{2\ell+1} & \text{if } j = 4\ell+2,\end{cases}$$

and

$$V^{-1}M_j = \begin{cases} M_{2\ell+1} & \text{if } j = 0,\\ M_{2\ell+2} & \text{if } j = 1,\\ M_{2\ell+3} & \text{if } 2 \leq j \leq 2\ell+1,\\ M_{j+2} & \text{if } 2k+1 \leq j \leq 4\ell,\\ M_{4\ell+2} & \text{if } 4k \leq j \leq 4\ell+2.\end{cases}$$

This shows that the M_j give the canonical filtration of M, and that the corresponding elementary sequence is $[0, \ldots, 0, 1, \ldots, p-a]$. Thus the a-number of M is a. For the classification by words, we note that the cycles corresponding to the filtration are

$$0 \xrightarrow{V^{-1}} 2\ell+1 \xrightarrow{V^{-1}} 2\ell+3 \xrightarrow{V^{-1}} \cdots \xrightarrow{V^{-1}} 4\ell+1 \xrightarrow{F} 2\ell \xrightarrow{F} 2\ell-2 \xrightarrow{F} \cdots \xrightarrow{F} 2 \xrightarrow{F} 0$$

(yielding the word $f^{\ell+1}v^{\ell+1}$) with multiplicity $b = \dim_k(M_1/M_0)$ and

$$1 \xrightarrow{V^{-1}} 2\ell+2 \xrightarrow{V^{-1}} 2\ell+4 \xrightarrow{V^{-1}} \cdots \xrightarrow{V^{-1}} 4\ell \xrightarrow{F} 2\ell-1 \xrightarrow{F} 2\ell-3 \xrightarrow{F} \cdots \xrightarrow{F} 3 \xrightarrow{F} 1$$

(yielding the word $f^\ell v^\ell$) with multiplicity $a - b = \dim_k(M_2/M_1)$.

The presentation by generators and relations then follows from [PU21, Lemma 3.1]. This completes the proof of the theorem. ☐

We can now give a significant application to Artin–Schreier covers:

Corollary 8.8. *Suppose that $p > 2$, $\pi : Y \to X$ and $G = \mathrm{Gal}(Y/X) \cong \mathbb{Z}/p\mathbb{Z}$ are as usual and that $f_X = g_X - 1$. Then the a-number of J_Y is in $\{2, 4, \dots, p-1, p\}$. Moreover, the Dieudonné module of $J_Y[p]$ has the form*

$$L \oplus (L \otimes k[G])^{f_X - 1} \oplus M_a,$$

where $L = M(\mathbb{Z}/p\mathbb{Z} \oplus \mu_p)$ and M_a is the module described in Theorem 8.7.

Proof. Since k is algebraically closed, $H^1_{dR}(Y)_{ét}$ is completely split, and part (2) of Proposition 4.1 gives its G-module structure. In all we have an isomorphism of $\mathbb{D}_k[G]$-modules

$$H^1_{dR}(Y)_{ét} \cong M(\mathbb{Z}/p\mathbb{Z}) \oplus (M(\mathbb{Z}/p\mathbb{Z}) \otimes_k k[G])^{f_X - 1}.$$

Similarly,

$$H^1_{dR}(Y)_m \cong M(\mu_p) \oplus (M(\mu_p) \otimes_k k[G])^{f_X - 1}.$$

Since $H^1_{dR}(Y)_{ll} = M$ has the properties enumerated in Proposition 8.1, it is isomorphic to the $\mathbb{D}_k[G]$-module M_a described in Theorem 8.7. This completes the proof of the corollary. ☐

Example 8.9. Let $p = 5$ and $k = \mathbb{F}_p$. The table below illustrates that all possibilities listed in Corollary 8.8 occur, with X hyperelliptic of degree 7 and genus $g_X = 3$. Each base curve X has $f_X = 2 = g_X - 1$, and $\nu_X = 1$ so the specified cover $Y \to X$ is the unique unramified $\mathbb{Z}/p\mathbb{Z}$-cover defined over k.

$X : y^2 =$	$Y : z^3 - z =$	a_Y
$x^7 - x^5 - 2x^3 - 2x^2 + x - 1$	$(-2x^4 - x^2 + 1)y + 2$	2
$-x^7 - x^6 - x^5 + x^4 + x^2 - 2x$	$(-x^4 - 2x^3 + 2x^2 + 1)y$	4
$2x^7 - 2x^5 + 2x^4 - x$	$(-x^9 + 2x^7 - 2x^6 - x^5 + 2x^4 - x + 1)y$	5

The argument underlying Theorem 8.7 has consequences for other $\mathbb{D}_k[G]$-modules M with $M/\delta M$ superspecial:

Theorem 8.10. *Suppose that $p > 2$ and let M be a $\mathbb{D}_k[G]$-module with properties (1)–(4) as in Proposition 8.1 (in particular, free over R of rank $2h$) and such that $M/\delta M$ is superspecial. Choose coordinates as in Proposition 8.4, and let $a(M)$ be the a-number of M.*

1. *If D has the form $\delta^\alpha U$ with $1 \le \alpha \le p$ and $U \in \mathrm{GL}_h(R)$, then M is the direct sum of h copies of the module in Theorem 8.7. I.e., writing $p = \ell\alpha + \beta$ with $0 \le \beta < \alpha$, M is the module associated to the words $f^{\ell+1}v^{\ell+1}$ with multiplicity $h\beta$ and $f^\ell v^\ell$ with multiplicity $h(\alpha - \beta)$. If $\alpha < p$, then $h\alpha$ is even.*
2. *If $a(M) = h$ (the minimum possible value), then M is associated to the word $f^p v^p$ with multiplicity h. In this case, h is even.*
3. *If h is odd, then $a(M) \ge h + 1$.*
4. *If $a(M) < p$, then $a(M)$ is even.*

Proof. We begin by observing that if

$$D = \sum_{i=\alpha}^{p-1} \delta^i D_i$$

with the $D_i \in M_h(k)$, then an argument generalizing that of Lemma 8.3 shows that the "leading matrix" D_α is symmetric when α is even and skew-symmetric when α is odd. We also note that

$$\operatorname{Ker} F \cap \operatorname{Ker} V = \operatorname{Ker} F\,|_{\operatorname{Im} F}\ ,$$

so

$$a(M) = \dim_k \operatorname{Ker}\left(D : R^h \to R^h\right). \tag{8.1}$$

(1) If D is a power of δ times an invertible matrix, the argument of Theorem 8.7 calculating the canonical filtration goes through essentially verbatim with $\begin{pmatrix} 1 \\ D \end{pmatrix}$ replaced by the column span of $\begin{pmatrix} I \\ D \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ replaced by $\begin{pmatrix} 0 \\ R^h \end{pmatrix}$. The canonical filtration of M has the same length and same permutation as in Theorem 8.7, but the subquotients are h times as large. For the parity assertion, the remarks above show that if $h\alpha$ is odd, then the leading matrix D_α is skew-symmetric and of odd size, so not invertible.

(2) The assumption that $M/\delta M$ is superspecial implies that $M[\delta]$ is also superspecial, so we have $a(M) \geq h$. By (8.1), we can have equality only if δ^2 does not divide D and the leading matrix D_1 is invertible. Thus we are in the case $\alpha = 1$ of part (1), $\beta = 0$, and $\ell = p$, so M is associated to $f^p v^p$ with multiplicity h, and h must be even.

(3) If h is odd, we cannot be in case (2), so $a(M) > h$.

(4) Equation (8.1) shows that as an element of R, $\det D \in \delta^{a(M)} R^\times$, in other words, $a(M)$, the length of $\operatorname{Ker} D$, is the "valuation" of $\det D$. (This is literally true only if we lift D to a valuation ring, say $k[[\delta]]$. Projecting back to $R = k[[\delta]]/(\delta^p)$ gives the statement here.) Since ${}^t\tilde{D} = D$, we have $\widetilde{\det D} = \det D$, so if $a(M) < p$, then $a(M)$ must be even by Lemma 8.3. ∎

Theorem 8.7 and the first half of Theorem 8.10 show that if h or a is small, the possibilities for M are severely restricted. We can also make such a deduction when a is large with respect to h:

Theorem 8.11. *Let M be a $\mathbb{D}_k[G]$-module satisfying the properties* (1)–(4) *of Proposition* 8.1.

1. *If $a(M) = ph$ (the maximum value), then M and $M/\delta M$ are superspecial.*
2. *If $a(M) = ph - 1$, then $M/\delta M$ is superspecial and there are exactly h possibilities for M as a $\mathbb{D}_k$-module which are indexed by an integer $1 \leq r \leq h$. The corresponding Ekedahl–Oort structure is $[0, \ldots, 0, 1, \ldots, 1]$ with $ph - r$ zeroes and r ones. The corresponding multiset of words is fv with multiplicity $2h-2r$ and the word $(vf)^{r-1} v^2 (fv)^{r-1} f^2$ with multiplicity one.*

Note that the Ekedahl–Oort structures appearing in part (2) are a small fraction (h vs. ph) of those with a-number 1, i.e., the R-structure on M is a significant constraint.

Proof. (1) By equation (8.1), M has a-number ph precisely when $D = 0$, and it follows (as in the proof of Theorem 8.7) that M and $M/\delta M$ are superspecial.

(2) By equation (8.1), M has a-number $ph - 1$ precisely when D has k-rank 1, in which case we must have $D = \delta^{p-1} D_{p-1}$ where $D_{p-1} \in M_h(k)$ has rank 1. It follows that $D \equiv 0 \pmod{\delta}$, i.e., $M/\delta M$ is superspecial.

Choose a vector $v \in k^h$ so that the column span of D_{p-1} is kv, and let r be defined by

$$r := \dim_k \left(kv + kv^{(p^2)} + kv^{(p^4)} + \cdots \right).$$

(Here we write $v^{(p^a)}$ for the result of raising each coordinate of v to the p^a-th power.) Note that $1 \leq r \leq h$. We are going to show that M has the asserted invariants by computing its canonical filtration.

We use the definitions and results of [Oor01] as reviewed in [PU21, §3-4]. We index certain elements of the canonical filtration of M by their dimensions. In particular,

$$M_{ph} = \operatorname{Ker} V = \operatorname{Im} F = \text{column span (over } R\text{) of } \begin{pmatrix} I \\ D \end{pmatrix}.$$

Let $\phi : \{0, \dots, 2ph\} \to \{0, \dots, ph\}$ be the function such that $FM_i = M_{\phi(i)}$. (To define ϕ on this domain, we need to extend the canonical filtration to a final filtration, but that will play no role in what follows.) To say that the a-number of M is 1 is precisely to say that $\phi(ph) = 1$. Let

$$M_1 = F^2 M = k \begin{pmatrix} \delta^{p-1} v^{(p)} \\ 0 \end{pmatrix}$$

and compute

$$\begin{aligned}
M_{ph+1} &= V^{-1} M_1 = M_{ph} + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^2)} \end{pmatrix}, \\
M_2 &= F M_{ph+1} = k \begin{pmatrix} \delta^{p-1} v^{(p)} \\ 0 \end{pmatrix} + k \begin{pmatrix} \delta^{p-1} v^{(p^3)} \\ 0 \end{pmatrix}, \\
M_{ph+2} &= V^{-1} M_2 = M_{ph} + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^2)} \end{pmatrix} + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^4)} \end{pmatrix}, \\
&\vdots \\
M_{ph+r} &= V^{-1} M_2 = M_{ph} + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^2)} \end{pmatrix} + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^4)} \end{pmatrix} + \cdots + k \begin{pmatrix} 0 \\ \delta^{p-1} v^{(p^{2r})} \end{pmatrix}.
\end{aligned}$$

Note that the definition of r implies that FM_{ph+r} has dimension r. We have thus established that $\phi(ph + i) = i + 1$ for $1 \leq i < r$ and $\phi(ph + r) = r$. The self-duality of M implies that $\phi(ph - i) = 1$ for $1 \leq i < r$ and $\phi(ph - r) = 0$. This completely determines the Ekedahl –Oort structure of M as $[0, \dots, 0, 1, \dots, 1]$ with $ph - r$ zeroes and r ones.

We leave it as an exercise for the reader to check that the corresponding set of words is fv with multiplicity $2h - 2r$ and $(vf)^{r-1} v^2 (fv)^{r-1} f^2$ with multiplicity one. ∎

We now turn to the case with $p = 2$ and $M/\delta M$ superspecial where we are able to give an almost complete analysis. Before stating the main result, we record a simple lemma.

Lemma 8.12. *Suppose $p = 2$ and let M be a finite-dimensional, local-local $\mathbb{D}_k$-module with a fixed alternating pairing $\langle \cdot, \cdot \rangle$ satisfying $\langle Fm, n\rangle = \langle m, Vn\rangle^p$ for all $m, n \in M$. In order to equip M with the structure of a $\mathbb{D}_k[G]$-module with the properties (1)–(4) of Proposition 8.1, it suffices to give a $\mathbb{D}_k$-submodule $N \subset M$ which is maximal isotropic for the pairing and an isomorphism $\phi : N^\vee \to N$ of $\mathbb{D}_k$-modules which is symmetric, i.e., with $\phi^\vee = \phi$.*

Proof. Since N is maximal isotropic, the pairing on M induces an isomorphism of $\mathbb{D}_k$-modules $M/N \cong N^\vee$. We define $\delta : M \to M$ as the composition

$$M \twoheadrightarrow M/N \cong N^\vee \xrightarrow{\phi} N \hookrightarrow M.$$

Clearly $\delta^2 = 0$, so M has the structure of an $R = k[G]$-module. If $m_1, \ldots, m_d$ is a basis of a complement of N in M, then the same set is a basis of M as an R-module, so M is R-free.

Since each of the maps in the composition defining δ is a map of $\mathbb{D}_k$-modules, so is δ, so M is a $\mathbb{D}_k$-module satisfying parts (1)–(2) of Proposition 8.1.

For part (3), the pairing and the R-action are compatible if and only if $\langle \delta m_1, m_2\rangle = \langle m_1, \delta m_2\rangle$ for all $m_1, m_2 \in M$. (Here $p = 2$, so $\tilde{\delta} = \delta$.) This holds because we chose ϕ to be symmetric.

For part (4), since M and N are self-dual,

$$\dim_k F(M) = \frac{1}{2}\dim_k(M) \quad \text{and} \quad \dim_k F(N) = \frac{1}{2}\dim_k(N) = \frac{1}{4}\dim_k(M).$$

Choose a basis $m_1, \ldots, m_{d/2}$ for a complement of $F(N)$ in $F(M)$. Then these elements span an R-module of k-dimension d. Since δ commutes with $\mathbb{D}_k$, it is contained in $F(M)$, and thus equal to $F(M)$ by dimension considerations. This shows that $\operatorname{Im} F$ is a free R-module and completes the proof that M is a $\mathbb{D}_k$-module with properties (1)–(4) of Proposition 8.1. □

Theorem 8.13. *Assume that $p = 2$, and define a $\mathbb{D}_k[G]$-module M to be "admissible" if M has the properties (1)–(4) of Proposition 8.1 and $M/\delta M$ is superspecial.*

(1) *The multiset of words attached to an admissible M consist of fv with even multiplicity together with a self-dual multiset of words each of which is a concatenation of subwords of the form*
$$(vf)^{e_2} v^2 (fv)^{e_1} f^2$$
with $e_1, e_2 \geq 0$.

(2) *Every multiset of words $\mathcal{W}$ as in part (1) is the set of words attached to an admissible M.*

(3) *The Ekedahl–Oort structure attached to an admissible M starts with h zeroes, i.e., it has the form*
$$[0, \ldots, 0, \psi_{h+1}, \ldots, \psi_{2h}].$$

(4) *Every Ekedahl–Oort structure $\Psi = [0, \ldots, 0, \psi_{h+1}, \ldots, \psi_{2h}]$ of length $2h$ which starts with h zeroes is the Ekedahl–Oort structure of an admissible M.*

Remark 8.14. We will prove part (2) of the theorem with an explicit construction of a module M with the asserted structures as $\mathbb{D}_k$- and $R = k[G]$-modules. We do not claim that any admissible M is isomorphic as $\mathbb{D}_k[G]$-module to one of those constructed in part (2). (This is the reason we describe our analysis as "almost complete.") In fact, the examples constructed for part (2) have a direct factor (as $\mathbb{D}_k[G]$-module) whose underlying $\mathbb{D}_k$-module is $M(fv)$ with even multiplicity. However, Example 8.15 after the proof shows that not every admissible module has this property.

Proof. (1) Let M be an admissible $\mathbb{D}_k[G]$-module, and choose coordinates as in Proposition 8.4. We will use these coordinates to compute certain elements of the canonical filtration of M.

Since $M/\delta M$ is superspecial, the matrix D in Proposition 8.4(3) has the form $D = \delta D_1$ where $D_1 \in M_h(k)$. For vectors or matrices with coordinates in k or subspaces of k^h, we use an exponent $^{(p^a)}$ to denote the result of raising each entry or coordinate to the power p^a. Viewing D_1 as an endomorphism of k^h by left multiplication, define subspaces

$$K = \operatorname{Ker} D_1 \cap \operatorname{Ker} D_1^{(p^2)} \cap \operatorname{Ker} D_1^{(p^4)} \cap \cdots$$

and

$$I = \operatorname{Im} D_1 + \operatorname{Im} D_1^{(p^2)} + \operatorname{Im} D_1^{(p^4)} + \cdots .$$

Since k^h is finite-dimensional, each of these subspaces is in fact the intersection or sum of finitely many of the displayed terms, and we have $K^{(p^2)} = K$ and $I^{(p^2)} = I$.

We let words on the alphabet $\{f, v\}$ act on submodules of M as in [PU21, §3]: $fN = F(N)$ and $vN = V^{-1}(N)$. Straightforward calculation (similar to that in the proof of Theorem 8.11(2)) shows that for sufficiently large n, $(vf)^n M$ is independent of n. We write $(vf)^\infty M$ for the common value, and we find

$$(vf)^\infty M = \begin{pmatrix} R^h \\ \delta R^h + K^{(p)} \end{pmatrix}.$$

Similarly,

$$(vf)^\infty 0 = \begin{pmatrix} R^h \\ \delta I \end{pmatrix},$$

$$(fv)^\infty M = \begin{pmatrix} \delta R^h + K \\ 0 \end{pmatrix},$$

and

$$(fv)^\infty 0 = \begin{pmatrix} \delta I^{(p)} \\ 0 \end{pmatrix}.$$

Note that f and v carry the quotients $((vf)^\infty M) / ((vf)^\infty 0)$ and $((fv)^\infty M) / ((fv)^\infty 0)$ isomorphically onto one another. It follows that the multiplicity of fv in the multiset of words attached to M is

$$\dim_k K + \operatorname{codim}_k I = 2 \dim_k K.$$

Set $e = \dim_k K$.

Choose a complement $J \subset k^h$ to I (i.e., $k^h = I \oplus J$) with $J^{(p^2)} = J$. Then we find that the subspace

$$N := \begin{pmatrix} 0 \\ K^{(p)} + \delta J \end{pmatrix} + \begin{pmatrix} \delta K + \delta J^{(p)} \\ 0 \end{pmatrix}$$

of M is a $\mathbb{D}_k$-submodule isomorphic to $M(fv)^{2e}$. (Note that it is not in general an R-submodule.) Moreover, it is visible that the image N in $M/\delta M$ is isomorphic to e copies of $M(fv)$.

By the Kraft classification of $\mathbb{D}_k$-modules, we have an isomorphism $M \cong N \oplus N'$ of $\mathbb{D}_k$-modules where N' has dimension $4h - 4e$ over k and where the multiset of words associated to N' is self-dual and consists of primitive words of length > 2 (i.e., the words f, v, and fv do not appear). Since the image of N in $M/\delta M$ has dimension $2e$ and is isomorphic to $M(fv)^e$, the image of N' in $M/\delta M$ must have dimension at least $2h - 2e$ and contain a subspace isomorphic to $M(fv)^{h-e}$. In particular, the $u_{1,1}$-number of N' is at least $h - e$.

We now check that for $p = 2$, a self-dual $\mathbb{D}_k$-module N' of dimension $4n$ and $u_{1,1}$-number $\geq n$ whose associated set of primitive words consists of words of length > 2 in fact has $u_{1,1}$-number equal to n and all of its associated words are concatenations of words of the form $(vf)^{e_2}v^2(fv)^{e_1}f^2$. To that end, recall the calculation of $u_{1,1}$-numbers in [PU24]: Given a word w, rotate and subdivide it so that it has the form $w = s_1 s_2 \cdots s_\ell$ where each s_i ends in f^2 and has no other appearances of f^2. A word that ends in f^2, has no other appearances of f^2, and contains v^2 can be written in the form $(vf)^{e_2}v^2 t_i (fv)^{e_1} f^2$ where t_i is either empty or it starts with f and ends with v^2. (This defines e_1 and e_2.) By [PU24, Prop.],

$$u(M(w)) + u(M(w)^\vee) = \sum_{i=1}^{\ell} u_i,$$

where

$$u_i = \begin{cases} 0 & \text{if } s_i \text{ has no appearances of } v^2, \\ (e_2 + 1) + (e_1 + 1) & \text{if } s_i = (vf)^{e_2}v^2 t_i (fv)^{e_1} f^2. \end{cases}$$

We find that

$$u(M(w)) + u(M(w)^\vee) \leq \frac{1}{4} \dim_k \left(M(w) \oplus M(w)^\vee\right)$$

with equality if and only if the second case above always holds and all of the t_i are empty. This establishes that the words associated to N' are concatenations of words of the form $(vf)^{e_2}v^2(fv)^{e_1}f^2$, which completes the proof of part (1) of the theorem.

(2) Suppose fv appears in $\mathcal{W}$ with even multiplicity d. Give $N := M(fv) \otimes_k R$ the structure of a $\mathbb{D}_k[G]$-module by letting $\mathbb{D}_k$ act on the first factor and R on the second. Choose an alternating pairing on $M(fv)$ and use it to identify $M(fv) \otimes \delta$ with the dual of $M(fv) \otimes 1$. Lemma 8.12 shows that N is admissible, as is $N^{d/2}$. To finish the proof, we construct an admissible module with multiset of words equal to those in $\mathcal{W}$ except fv. Taking the direct sum with $N^{d/2}$ then provides the required module.

We thus may assume that $\mathcal{W}$ does not contain fv, so consists entirely of words which are concatenations of words of the form $(vf)^{e_2}v^2(fv)^{e_1}f^2$ with $e_1, e_2 \geq 0$. Let M be the $\mathbb{D}_k$-module associated to the words in $\mathcal{W}$ and let $4h = \dim_k M$, so $4h$ is also the number of letters in $\mathcal{W}$. We claim that there is a surjection of

$\mathbb{D}_k$-modules $M \twoheadrightarrow M(fv)^h$ whose kernel is isomorphic as $\mathbb{D}_k$-module to $M(fv)^h$. Choosing a symmetric pairing on $M(fv)^h$ and applying Lemma 8.12, we see that M admits a $\mathbb{D}_k[G]$-module structure which makes it admissible.

The proof of [PU24, Proposition] shows that each subword $(vf)^{e_2}v^2(fv)^{e_1}f^2$ leads to a (non-unique) surjection $M \twoheadrightarrow M(fv)^{e_2+1}$ and (dually) a (non-unique) injection $M(fv)^{e_1+1} \hookrightarrow M$. With appropriate choices, these fit together to make M into an extension of $M(fv)^h$ by itself. We explain this for a single subword and leave the mild generalization to concatenations to the reader.

Let $w = (vf)^{e_2}v^2(fv)^{e_1}f^2$. Present the module $M(w)$ as in [PU21, §3.2] with generators $E_0, E_1, \ldots, E_{e_1+e_2+1} = E_0$ and relations

$$VE_{e_1+e_2+1} = FE_{e_1+e_2},$$
$$\vdots$$
$$V^2E_{e_1+1} = FE_{e_1},$$
$$\vdots$$
$$VE_1 = F^2E_0.$$

(For notational convenience we have assumed $e_1, e_2 > 0$. The other cases are similar but simpler.) We write $z_0^{(j)}$ and $z_1^{(j)}$ (with $j = 1, \ldots, e_2+1$) for bases of copies of $M(fv)$ where $Fz_0^{(j)} = Vz_0^{(j)} = z_1^{(j)}$ and $Fz_1^{(j)} = Vz_1^{(j)} = 0$. A surjection to $M(fv)^{e_2+1}$ is given by

$$E_i \mapsto \begin{cases} 0 & 1 \le i \le e_1, \\ \alpha_j^{p^{2(i-e_1-1)}} z_0^{(i-e_1)} & e_1+1 \le i \le e_1+e_2+1, \end{cases}$$

where $\alpha_1, \ldots, \alpha_{e_2+1} \in k$ are chosen so that the matrix

$$\begin{pmatrix} \alpha_1 & \alpha_1^{p^2} & \ldots & \alpha_1^{p^{2e_2}} \\ \vdots & \vdots & & \vdots \\ \alpha_{e_2+1} & \alpha_{e_2+1}^{p^2} & \ldots & \alpha_{e_2+1}^{p^{2e_2}} \end{pmatrix}$$

has rank e_2+1. Let $\beta_0, \beta_1, \ldots, \beta_{e_2+1}$ be a non-zero solution to the system

$$\begin{pmatrix} \alpha_1^{1/p} & \alpha_1^p & \alpha_1^{p^3} & \ldots & \alpha_1^{p^{2e_2+1}} \\ \vdots & \vdots & \vdots & & \vdots \\ \alpha_{e_2+1}^{1/p} & \alpha_{e_2+1}^p & \alpha_{e_2+1}^{p^3} & \ldots & \alpha_{e_2+1}^{p^{2e_2+1}} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{e_2+1} \end{pmatrix} = 0,$$

and note that by our choice of the α_i, we have that β_0 and β_{e_2+1} are non-zero. One then checks that the kernel of the surjection $M \twoheadrightarrow M(fv)^h$ is the k-vector space N spanned by

$$\beta_0VE_{e_1+1} + \beta_1FE_{e_1+1} + \cdots + \beta_{e_2+1}FE_{e_1+e_2+1}$$

and

$$E_1, \ldots, E_{e_1}, F^2E_0, FE_1, \ldots FE_{e_1}.$$

Let N_1 be the subspace spanned by the first e_1+1 vectors listed above. Then one finds that $FN = VN = N_1$ and $FN_1 = VN_1 = 0$. This shows that N is isomorphic as $\mathbb{D}_k$-module to $M(fv)^{e_1+1}$.

Making a similar analysis of concatenations of words of the form $(vf)^{e_2}v^2(fv)^{e_1}f^2$ shows that M is an extension of $M(fv)^h$ by $M(fv)^h$, and this completes the proof of part (2) of the theorem.

For part (3), assume that M is associated to a set of cyclic words consisting of fv with multiplicity $2d$ and concatenations of words

$$s_i = (vf)^{e_{2i}}v^2(fv)^{e_{2i-1}}f^2 \qquad i = 1, \dots, m$$

(with repetitions as necessary to account for multiplicities). Note that self-duality implies that $\sum_{i=1}^m e_{2i} = \sum_{i=1}^m e_{2i-1}$. Let $h = d + \sum_{i=1}^m e_{2i}$ so that M has dimension $4h$ over k.

A basis of M is indexed by the words obtained by lifting each cyclic word above to a word in all possible rotations. In the standard model of $M(w)$, a basis vector associated to a word ending in f is mapped by F to the basis vector corresponding to the first rotation of the word, and a basis vector corresponding to a word ending with v is killed by F. (See [PU21, 3.2.1] for more details.)

We recall from [PU21, §3.5] a total order on words on $\{f, v\}$: given two words w_1 and w_2, form powers w_1^a and w_2^b of the same length, and then compare them in lexicographic order (from the left with $f < v$). The canonical filtration of M is obtained by taking subspaces generated by the vectors associated to words less than or equal to one of the words which appears (with repetitions for multiplicities).

We now consider which words in $\mathcal{W}$ are less than or equal to fv. They are those which start with $f^2\dots$ or with $(fv)^j f^2 \dots$ with $1 \le j \le e_{2i-1}$ and $1 \le i \le m$ as well as fv itself (taken with multiplicities). Note that all of these words end in v. The total number of them is $2d + \sum_{i=1}^m e_{2i-1} = h + d$. Since words that end in v are killed by F, we see that the subspace M_{d+h} in the canonical filtration of M with dimension $d + h$ is killed by F, and this means that the Ekedahl–Oort structure associated to M begins with $h + d$ zeroes. This establishes the claim in part (3) of the theorem.

(4) We consider the $\mathbb{D}_k$-module M with E–O structure Ψ as constructed by Oort in [Oor01, §9] and show that M admits a $k[G]$-module structure which makes it admissible.

Extend Ψ to a "final sequence" $[\psi_1, \dots, \psi_{4h}]$ by setting $\psi_{4h} = 2h$ and $\psi_{4h-i} = \psi_i + 2h - i$ for $1 \le i \le 2h$. Let $1 \le m_1 < m_2 < \cdots < m_{2h} \le 4h$ be the indices i such that $\psi_{i-1} < \psi_i$ and let $1 \le n_{2h} < n_{2h-1} < \cdots n_1 \le 4h$ be the indices i such that $\psi_{i-1} = \psi_i$. Our hypothesis on Ψ implies that $m_{h+i} = 3h + i$ for $1 \le i \le h$ and $n_{2h+1-i} = i$ for $1 \le i \le h$. Moreover, $m_i + n_i = 4h + 1$ for $1 \le i \le 2h$.

Now let M be the k-vector space with basis $Z_1, \dots, Z_{4h}$ and for $1 \le i \le 2h$ define $X_i = Z_{m_i}$ and $Y_i = Z_{n_i}$. Define a $\mathbb{D}_k$ module structure on M by setting

$$F(X_i) = Z_i, \quad F(Y_i) = 0, \quad V(Z_i) = 0, \quad \text{and} \quad V(Z_{4h+1-i}) = Y_i \quad \text{for } 1 \le i \le 2h.$$

Introduce a bilinear pairing $\langle \cdot, \cdot \rangle$ on M by setting

$$\langle X_i, X_j \rangle = 0, \quad \langle Y_i, Y_j \rangle = 0, \quad \text{and} \quad \langle X_i, Y_j \rangle = \delta_{ij} \quad \text{for } 1 \le i, j \le 2h.$$

It is then straightforward to check that M is a self-dual, local-local BT_1 module whose elementary sequence is the given Ψ.

Let N be the subspace of M spanned by $X_i + Y_i$ and Y_{h+i} for $1 \le i \le h$. Then N is a $\mathbb{D}_k$-submodule and we have

$$F(N) = V(N) = \text{Span of } Y_{h+i} \text{ for } 1 \le i \le h.$$

Since $F(Y_{h+i}) = V(Y_{h+i}) = 0$ for $1 \le i \le h$, it follows that N is superspecial.

The quotient M/N is spanned by the classes of X_j for $1 \le j \le 2h$, and we have

$$F(M/N) = V(M/N) = \text{Span of the classes of } X_i \text{ for } 1 \le i \le h.$$

Since $F(X_i) = V(X_i) = 0 \pmod{N}$ for $1 \le i \le h$, it follows that $M/N \cong N^\vee$ is also superspecial. Choosing a symmetric isomorphism $N^\vee \to N$ and applying Lemma 8.12, we find the desired R-module structure on M.

This completes the proof of the theorem. □

Example 8.15. Let $p = 2$ and consider the $\mathbb{D}_k[G]$-module M of rank $2h = 4$ over R given as in Proposition 8.4(4) by $D = \delta D_1$ with

$$D_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Note that $D_1^2 = 0$. The Ekedahl–Oort structure associated to M is $[0,0,0,1]$ and the associated set of words is $(f^2v^2), (fv)^2$. The submodule $M_7 := V^{-1}FM$ has dimension 7, and one checks that if $m \in M \setminus M_7$, then the $\mathbb{D}_k[G]$-submodule of M generated by m has dimension 6 and is isomorphic as R-module to $R^2 + \delta R^2$. In particular, the smallest free R-submodule of M containing it is all of M. This proves that M is indecomposable as a $\mathbb{D}_k[G]$-module. Note by contrast that the $\mathbb{D}_k$-module associated to $(f^2v^2), (fv)^2$ in part (2) of Theorem 8.13 decomposes into a free R-module of rank 2 supporting the $\mathbb{D}_k$-module $(fv)^2$ and another of rank 2 supporting (f^2v^2). This shows that not every admissible $\mathbb{D}_k[G]$-module arises from the construction in part (2).

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